Kaye Validator® AVS

Installation Qualification Protocol
Version 1.5.1

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Validation :	_____	Date :	_____
Subject Matter Expert :	_____	Date :	_____
Management :	_____	Date :	_____
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1. PURPOSE

The purpose of this document is to outline the test plan and documentation requirements that will be used to qualify the installation of the Kaye Validator AVS system. The Installation Qualification (IQ) procedures in this Validation Protocol are intended to ensure that the Kaye Validator AVS and its associated hardware and software components are properly installed according to the recommendations of Amphenol Advanced Sensors, Inc., and are adequately documented and controlled according to cGMP requirements.

2. EQUIPMENT DESCRIPTION

The Kaye Validator AVS system is in Amphenol Advanced Sensors line of thermal validation systems. The Kaye Validator AVS system consists out of the Kaye Validator AVS instrument and the Kaye Validation Console (HMI, Human Machine Interfaces) and is a compact, modular, and portable instrument designed specifically to meet current thermal validation needs and regulations. The Kaye Validator AVS can accommodate up to 48 sensor inputs, with any combination of thermocouple, voltage, current or RTD inputs. The Kaye Validator AVS is designed to connect easily to Amphenol Advanced Sensors temperature standards and reference baths. This modular design allows for fully automatic sensor calibration, as well as multiple calibration or qualification equipment configurations.

Sensor input is accomplished through a Sensor Input Module (SIM), a modular sensor input interface that protects the electronics against dust, humidity, electrical noise, and mechanical shock. Each SIM has 12 sensor input connections, a memory chip to store calibration offsets, SIM serial number, calibration slot location, and the serial number of the Kaye Validator AVS measurement board where calibration was performed, and a cold junction reference RTD to help maintain measurement accuracy through changing environmental conditions. The temperature measurement range is -270°C to 1800°C, depending on the type of thermocouple used, with a total system measurement accuracy of 0.28°C at 121.0°C.

The Kaye Validator AVS system Software is pre-installed and configured on the dedicated Kaye Validation Console and provides everything needed to perform validation testing. Using the Kaye Validator AVS Software, a System Administrator creates user accounts and assigns user IDs, passwords, and access permission levels that identify the person operating the instrument and prevent unauthorized operation. Validation Assets containing study setups, which include calibration criteria, are programmed using the Kaye Validator AVS Software. The Kaye Validator AVS also provides for encoded, secure data file storage that prevents modification of the original data, and includes extensive, configurable data reporting options, either as printed hardcopies or spreadsheet-compatible electronic files. This secure electronic data format is designed for implementation of 21 CFR Part 11 requirements.

The Kaye Validator AVS has 4GB customer accessible on-board memory, USB forced download of last 10 studies and a 32GB SD card (internal). The Kaye Validator AVS instrument is designed to write data to internal memory and the Kaye Validation Console hard drive (if connected) during sensor Calibration, Calibration Verification, and Qualification studies. Qualification data resides in internal memory until the Kaye Validation Console is connected to the Kaye Validator AVS instrument or data is downloaded to a USB flash drive (last 10 studies). Connection can be established via dock-in, Ethernet or Wi-Fi. The Kaye Validation Console must be used to transfer Setups, Calibration data, and Qualification data between the Kaye Validation Console equipped with Kaye Validator AVS Software and the Kaye Validator AVS instrument.

The Kaye Validator AVS is equipped with a lithium ion backup battery to provide short-term power in case of AC power loss. The intent of the battery is to provide sufficient time (minimum of 3hrs) so that you can perform an orderly shutdown of the system. When the Kaye Validator AVS detects that the AC power has failed and the system is running from battery, it displays an icon on the front panel. When running on battery, the Kaye Validator AVS instrument monitors its own battery voltage. When the battery has approximately 6 minutes of power remaining, the Kaye Validator AVS instrument stops collecting data and closes open data files to prevent data loss.

The Kaye Validation Console comes preconfigured with the Kaye Validator AVS Software installed. At least one Kaye Validation Console is necessary to operate the Kaye Validator AVS instrument but it is also possible to use more than one Kaye Validation Console to operate the Kaye Validator AVS instrument or to check system status.

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The Kaye Validator AVS is designed to operate with the Kaye IRTD 400 (a self-contained, high-accuracy temperature standard) and a Kaye Temperature Reference (LTR-90, LTR –40/140, LTR –25/140, LTR-150, HTR 400, HTR-420). The IRTD 400 provides NIST or DAkkS traceable calibration of thermocouples used in thermal validation studies, with a range of -195°C to 420°C, and an accuracy around of 0.030°C. During Calibration and/or Qualification, the Kaye Validator AVS automatically reads and monitors the temperature of the IRTD and the Temperature Reference. For thermocouple calibration, the IRTD functions as the standard for correcting thermocouple readings. The Temperature Reference provides a highly stable temperature environment for sensor calibration, and responds to programmed calibration criteria, allowing automatic thermocouple calibration at both low- and high-temperature setpoints, with a third setpoint for a mid-range temperature check.

3. QUALIFICATION METHODOLOGY

Complete validation of the Kaye Validator AVS system will be demonstrated by the successful completion of both Installation and Operational Qualification Protocols. The Installation Qualification will be performed in the execution of this Validation Protocol. The Operational Qualification will be performed in a separate Validation Protocol.

This Installation Qualification is intended to verify and document that the Kaye Validator AVS system and its associated hardware and software components are properly installed according to the recommendations of Amphenol Advanced Sensors, and adequately documented and controlled according to cGMP requirements. The Acceptance Criteria for each IQ procedure have been derived from the manufacturer's recommendations and/or cGMP requirements.

The purpose, methodology, and derivation of the Acceptance Criteria of each IQ procedure of this Validation Protocol are discussed below.

Note: The Kaye Validator AVS system is a portable instrument intended to be used throughout the facility on a routine basis. Although Amphenol Advanced Sensors specifies an operating environment of 0° to 50°C and a relative humidity of <95% (non-condensing), as well as a nominal supply voltage of 115 VAC or 230VAC, these environmental specifications will not be verified in this Installation Qualification because the instrument will not be permanently installed in a single location.

3.1. Vendor Documentation Verification (Section 6.1.)

The purpose of this IQ procedure is to inventory the documentation items provided by Amphenol Advanced Sensors with the Kaye Validator AVS system. This will be accomplished by verifying the identity, receipt, and location of each of the specified items, which include documentation for the basic Kaye Validator AVS instrument, as well as optional equipment and accessories. The Acceptance Criteria for this section are derived from general cGMP equipment or instrument documentation requirements.

3.2. Customer Documentation Verification (Section 6.2.)

The purpose of this IQ procedure is to inventory the documentation items listed from the equipment shipping/receiving documents. This will be accomplished by verifying the identity, receipt, and location of each of the specified items associated with the purchase and receipt of the equipment. Also included in this IQ procedure is a listing of Standard Operating Procedures (SOPs) that address the different functions of the Kaye Validator AVS system. The Acceptance Criteria for this section are derived from general cGMP equipment or instrument documentation requirements.

3.3. Hardware Information (Section 6.3.)

The purpose of this IQ procedure is to record the basic hardware information for primary and accessory components of the Kaye Validator AVS system. This will be accomplished by recording the model, serial number, and internal ID number of each of the hardware items associated with the Kaye Validator AVS system. Also included in this IQ procedure is a verification of the supporting components of the Kaye Validator AVS system. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations in the *Kaye Validator AVS User's Guide*, and general cGMP equipment or instrument documentation requirements.

3.4. Software/Firmware Information (Section 6.4.)

The purpose of this IQ procedure is to record the critical software and firmware information for the Kaye Validator AVS System. This will be accomplished by recording the version number and the Release Date (if available) of

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the software and firmware associated with the Kaye Validator AVS System. The Acceptance Criteria for this section are derived from cGMP requirements for software identification.

3.5. Calibration Verification (Section 6.5.)

The purpose of this IQ procedure is to verify the receipt of accurate calibration documentation provided by Amphenol Advanced Sensors for the relevant Kaye Validator AVS system components. This will be accomplished by documenting and verifying the receipt and accuracy of the calibration record of each specified component. The Acceptance Criteria for this section are derived from general cGMP equipment or instrument documentation requirements. The cGMP requirements for routine calibration tracking and control will be addressed in a separate Qualification procedure.

4. RESPONSIBILITIES

Insert the names/titles/departments of all persons responsible for each of the following activities listed in Table 4.1.

Table 4.1. Responsibility Record

Activity	Person Responsible	Position/Title	Department
Protocol Execution			
Protocol Execution Verification			
Discrepancy Review and Approval			
Report Generation			
Report Approval			

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5. EXECUTION REQUIREMENTS

Record all required data on the appropriate data sheets as it is collected. Record explanatory notes in the **Comments** section of the appropriate data sheets as necessary. Transcription errors may be corrected by crossing out the error with a single line, recording the corrected information, and initialing and dating the correction according to cGMP documentation requirements.

This Validation Protocol has been designed to account for a complete Kaye Validator AVS system. In some cases, the IQ procedures refer to equipment, documentation, or other items beyond the scope of the system being validated. Verification items relating to optional Kaye Validator AVS system equipment and accessories that were not purchased or are not included with the Kaye Validator AVS system should be labeled "**Not Purchased**" or "**N/P**" to indicate that the equipment or accessory requires no documentation. Verification items that are extraneous or not relevant should be labeled "**Not Applicable**" or "**N/A**" to indicate that the specification is not relevant to the item being documented or verified. Verification items that are not currently available should be labeled "**Not Available**" or "**N/AV**".

Supporting data or documentation to be included with the executed Protocol and Final Report must be labeled with the Protocol number, a reference to the specific IQ procedure and/or Protocol page that the data supports, and the initials and date of the person who collected the supporting data.

If the supporting data or documentation is to be stored to CD or other media, the files will be named to identify the purpose of the file and/or the specific IQ procedure it supports. The CD will be labeled and have the initials and date of the person who collected and stored the supporting data.

When a Discrepancy is encountered during the execution of this Protocol, complete a copy of the Validation Discrepancy Report form (see **Section 7**) according to the corresponding IQ procedure. Make a note in the **Comments** section of the corresponding IQ procedure, including the Discrepancy Number. The Discrepancy must be documented, resolved, and approved by the responsible reviewers before the execution of this Protocol can be considered complete and the Final Report can be approved.

Submit completed IQ procedures to the responsible reviewer on a timely basis.

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5.1. Signature Record

Each person involved in documenting, verifying, or approving any part of the execution of this Protocol (including data collection and Discrepancies) must record the specified information in Table 5.1.1. in order to positively identify all signatures and initials recorded in the execution of this Protocol. If a system for signature and initial identification is already in place, this signature record need not be completed.

Table 5.1.1. Signature Record

Printed Name	Signature	Initials	Position/Title	Date

5.2. Materials and Equipment

- Kaye Validator AVS instrument with Firmware Version 1.5
- At least one Kaye Validation Console with Software Version 1.5.1
- Kaye Sensor Input Module(s) (SIM)
- Kaye Temperature Reference(s)
- Kaye IRTD(s)
- Equipment User's Guides/Manuals
- Kaye Validator AVS Validation Reference
- Various documentation related to the Kaye Validator AVS system's order, delivery, operation, calibration and maintenance

5.3. Training

Persons participating in the execution of this Validation Protocol must be adequately trained. Training requirements should include:

- Operation of the Kaye Validator AVS system and associated equipment
- Good Documentation Practices

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6. QUALIFICATION PROCEDURES

Complete each of the qualification procedures described on the following pages according to the specified procedures and all applicable execution requirements listed in Section 5. Complete each data table and verify that the collected data conforms to the Acceptance Criteria for the corresponding procedure.

6.1. Vendor Documentation Verification

Procedure:

1. Record the information specified for each documentation item listed in Tables 6.1.1. and 6.1.2.

Table 6.1.1. Kaye Validator AVS System Documentation

Document Description	Part Number	Revision Number	Document Location	Initials/Date
Kaye Validator AVS User's Manual	M5100-EN			
Kaye Validator AVS Standard Operating Procedure	M4955			
Validation Reference **Note:** This is an optional document, which can be purchased.	X6050			

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Table 6.1.2. Kaye Validator AVS System Accessories Documentation

Document Description	Part Number	Revision Number	Document Location	Initials/Date
IRTD Manual	M2845-6			
LTR-90 Manual	M4590			
LTR –40/140 Manual	M4374			
LTR –25/140 Manual	M4374			
LTR-150 Manual	M5101			
HTR 400 Manual	M4395			
HTR-420 Manual	M4395			

Comments:

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_____

_____

Vendor Documentation Verification (Section 6.1.)

Acceptance Criteria:

All vendor documentation items for equipment included with the Kaye Validator AVS system, as well as optional equipment and accessories, have been located, listed, and verified. The Acceptance Criteria for this section are derived from general cGMP equipment or instrument documentation requirements.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.2. Customer Documentation Verification

Procedure:

1. Record the information specified for each documentation item listed in Table 6.2.1.
2. Fill in additional Document Descriptions and the corresponding specified information for any additional documents.
3. Verify that the documentation accurately corresponds to the equipment received.
4. Identify any SOPs that address system function such as those listed in Table 6.2.2., and record the specified information.

Table 6.2.1. Purchase and Receipt Documentation

Document Description	Document Number	Date	Document Available? (Y/N)	Initials/Date
Quotation				
Purchase Order				
Shipping/Receiving Document				

Table 6.2.2. Standard Operating Procedures

System Function	Document Number	Effective Date	Document Available? (Y/N)	Initials/Date
Kaye Validator AVS System Administration				
Asset Programming				
Sensor Calibration				
Performing Qualifications				
Report Printing				
Data Storage and Backup				
Kaye Validator AVS instrument Calibration				

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Comments:

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Customer Documentation Verification (Section 6.2.)

Acceptance Criteria:

All customer documentation items for equipment included with the Kaye Validator AVS system have been located, listed, and verified. All Kaye Validator AVS system functions are addressed, and the relevant SOPs have been identified. The Acceptance Criteria for this section are derived from general cGMP equipment or instrument documentation requirements.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.3. Hardware Information

Procedure:

1. Record the information of equipment purchased as well as existing equipment to be used for Kaye Validator AVS IQ/OQ execution.

Table 6.3.1. Hardware Information

Hardware Component	Model/Type	Serial Number	Internal Asset ID	Location	Initials/Date
Kaye Validator AVS instrument	X2015				
Kaye Validation Console	X2604				
Sensor Input Module (SIM) #1					
Sensor Input Module (SIM) #2					
Sensor Input Module (SIM) #3					
Sensor Input Module (SIM) #4					
Sensor Input Module (SIM) #5					
Sensor Input Module (SIM) #6					
Sensor Input Module (SIM) #7					
Sensor Input Module (SIM) #8					
IRTD #1					
IRTD #2					
LTR-90	X0395				
LTR –40/140	X0358				
LTR –25/140	X0356				
LTR-150	X0365				
HTR 400	X0361				
HTR-420	X0358				

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Table 6.3.2. Supporting Components Information

Supporting Component	Received? (Y/N)	Initials/Date
One (1) AC Power Cord for the Kaye Validator AVS instrument		
One (1) Battery pack for the Kaye Validator AVS instrument		
Two (2) Output Relay Connectors (N-C/N-O)		
IRTD Connection Cable(s)		
Temperature Reference Connection Cable(s)		
One (1) Power Supply for the Kaye Validation Console		
One (1) Battery pack for the Kaye Validation Console		
One (1) Touch Pen for the Kaye Validation Console		

Comments:

_____

_____

_____

_____

Hardware Information (Section 6.3.)

Acceptance Criteria:

All specified hardware information has been recorded. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations in the *Kaye Validator AVS User's Guide*, and general cGMP equipment or instrument documentation requirements.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.4. Software/Firmware Information

Procedure:

1. Login to the Kaye Validator AVS software. Select **Discover** below Hardware.
2. Select **USB/Docking** and then select **Connect** at the bottom right of the screen. Verify that the top Communication LED on the front of the Kaye Validator AVS is green, this will signalize communication between Kaye Validator AVS instrument and the Kaye Validation Console.
3. By swiping your finger down on the top of the Touchscreen a Popup menu will appear.
4. Select **About.** Record the information specified for the Kaye Validator AVS system Software components listed in Table 6.4.1.
5. Login to the Kaye Validator AVS Reporting Tool software.
6. By swiping your finger down on the top of the Touchscreen a Popup menu will appear.
7. Select **About.** Record the information specified for the Kaye Validator AVS Reporting Tool Software component listed in Table 6.4.1.

Table 6.4.1. Software/Firmware Information

Software/Firmware Component	Version	Initials/Date
Kaye Validator AVS System Software (Software Version)		
Kaye Validator AVS System Firmware (Firmware Version)		
Kaye Validator AVS Reporting Tool Software (Software Version)		

Comments:

_____

_____

_____

_____

Software/Firmware Information (Section 6.4.)

Acceptance Criteria:

The specified information for the software and firmware associated with the Kaye Validator AVS system and the Kaye Validator AVS Reporting Tool Software has been recorded. The Acceptance Criteria for this section are derived from cGMP requirements for software identification.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.5. Calibration Verification

Procedure:

1. Verify that a Calibration Record (CR) addressing each component of the Kaye Validator AVS system has been received.
2. Verify that the corresponding Calibration Record matches each component of the Kaye Validator AVS system and is accurate (model, serial, and date of calibration all correspond).
3. Record the calibration date of each component according to the Calibration Record.

Table 6.5.1. Calibration Information

Hardware Component	Serial No.:	Calibration Record Received? (Y, N, N/A)	Calibration Record Accurate? (Y, N, N/A)	Calibration Date(s) (DD-MMM-YYYY)	Initials/Date
Kaye Validator AVS instrument	Refer to Table 6.3.1				
Sensor Input Modules (SIM) #1					
Sensor Input Modules (SIM) #2					
Sensor Input Modules (SIM) #3					
Sensor Input Modules (SIM) #4					
Sensor Input Modules (SIM) #5					
Sensor Input Modules (SIM) #6					
Sensor Input Modules (SIM) #7					
Sensor Input Modules (SIM) #8					
IRTD #1					
IRTD #2					
LTR-90					
LTR –40/140					
LTR –25/140					
LTR-150					
HTR 400					
HTR-420					

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Comments:

_____

_____

_____

_____

Calibration Verification (Section 6.5.)

Acceptance Criteria:

An accurate calibration record has been received for each component of the Kaye Validator AVS. The Acceptance Criteria for this section are derived from general cGMP equipment or instrument documentation requirements.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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7. DISCREPANCY REPORT

Procedure:

1. Assign a sequential identification number to the Discrepancy and record the Protocol section or page number that the Discrepancy references.
2. Document a complete description of the Discrepancy, specifically addressing how it impacts the Acceptance Criteria of the Validation procedure. Sign and date the Discrepancy description and notify the designated reviewer.
3. The responsible personnel must determine a suitable Resolution to the Discrepancy and document the planned Resolution on the Discrepancy Report.
4. The designated reviewer must verify that the Resolution has been executed satisfactorily, sign, and date the Discrepancy Report.
5. Submit the Discrepancy Report to the designated QA reviewer for approval. Include the approved Discrepancy Report as Supporting Documentation for the affected Validation procedure.

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7. DISCREPANCY REPORT (continued)

Discrepancy Report Form
(Copy this form as needed) Page _____ of _____ .

Discrepancy No.:		**Protocol Section:**		**Page No.:**	
Validation Discrepancy Description:					
Acceptance Criteria Impact:					
Submitted By:				**Date:**	
Validation Discrepancy Resolution:					
Recorded By:				**Date:**	
Reviewed By:				**Date:**	
QA Approval:				**Date:**	

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Kaye Validator® AVS

Operational Qualification Protocol
Version 1.5.1

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Approvals (Name and Signature)			
System Owner :	_____	Date :	_____
Validation :	_____	Date :	_____
Subject Matter Expert :	_____	Date :	_____
Management :	_____	Date :	_____
Quality/Regulation :	_____	Date :	_____

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1. PURPOSE

The purpose of this document is to outline the test plan and documentation requirements that will be used to qualify the operation of the Kaye Validator AVS system. The Operational Qualification (OQ) procedures in this Validation Protocol are intended to ensure that the Kaye Validator AVS and its associated hardware and software components operate as specified according to the recommendations of Amphenol Advanced Sensors, Inc., and are adequately documented and controlled according to cGMP requirements.

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Sensor input is accomplished through a Sensor Input Module (SIM), a modular sensor input interface that protects the electronics against dust, humidity, electrical noise, and mechanical shock. Each SIM has 12 sensor input connections, a memory chip to store calibration offsets, SIM serial number, calibration slot location, and the serial number of the Kaye Validator AVS measurement board where calibration was performed, and a cold junction reference RTD to help maintain measurement accuracy through changing environmental conditions. The temperature measurement range is -270°C to 1800°C, depending on the type of thermocouple used, with a total system measurement accuracy of 0.28°C at 121.0°C.

The Kaye Validator AVS system Software is pre-installed and configured on the dedicated Kaye Validation Console and provides everything needed to perform validation testing. Using the Kaye Validator AVS Software, a System Administrator creates user accounts and assigns user IDs, passwords, and access permission levels that identify the person operating the instrument and prevent unauthorized operation. Validation Assets containing study setups, which include calibration criteria, are programmed using the Kaye Validator AVS Software. The Kaye Validator AVS also provides for encoded, secure data file storage that prevents modification of the original data, and includes extensive, configurable data reporting options, either as printed hardcopies or spreadsheet-compatible electronic files. This secure electronic data format is designed for implementation of 21 CFR Part 11 requirements.

The Kaye Validator AVS has 4GB customer accessible on-board memory, USB forced download of last 10 studies and a 32GB SD card (internal). The Kaye Validator AVS instrument is designed to write data to internal memory and the Kaye Validation Console hard drive (if connected) during sensor Calibration, Calibration Verification, and Qualification studies. Qualification data resides in internal memory until the Kaye Validation Console is connected to the Kaye Validator AVS instrument or data is downloaded to a USB flash drive (last 10 studies). Connection can be established via dock-in, Ethernet or Wi-Fi. The Kaye Validation Console must be used to transfer Setups, Calibration data, and Qualification data between the Kaye Validation Console equipped with Kaye Validator AVS Software and the Kaye Validator AVS instrument.

The Kaye Validator AVS is equipped with a lithium ion backup battery to provide short-term power in case of AC power loss. The intent of the battery is to provide sufficient time (minimum of 3hrs) so that you can perform an orderly shutdown of the system. When the Kaye Validator AVS detects that the AC power has failed, and the system is running from battery, it displays an icon on the front panel. When running on battery, the Kaye Validator AVS monitors its own battery voltage. When the battery has approximately 6 minutes of power remaining, the Kaye Validator AVS stops collecting data and closes open data files to prevent data loss.

The Kaye Validation Console comes preconfigured with the Kaye Validator AVS Software installed. At least one Kaye Validation Console is necessary to operate the Kaye Validator AVS instrument, but it is also possible to use more than one Kaye Validation Console to operate the Kaye Validator AVS instrument or to check system status.

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The Kaye Validator AVS is designed to operate with the Kaye IRTD 400 (a self-contained, high-accuracy temperature standard) and a Kaye Temperature Reference (LTR-90, LTR –40/140, LTR –25/140, LTR-150, HTR 400, HTR-420). The IRTD 400 provides NIST or DAkkS traceable calibration of thermocouples used in thermal validation studies, with a range of -195°C to 420°C, and an accuracy around of 0.030°C. During Calibration and/or Qualification, the Kaye Validator AVS automatically reads and monitors the temperature of the IRTD and the Temperature Reference. For thermocouple calibration, the IRTD functions as the standard for correcting thermocouple readings. The Temperature Reference provides a highly stable temperature environment for sensor calibration, and responds to programmed calibration criteria, allowing automatic thermocouple calibration at both low- and high-temperature setpoints, with a third setpoint for a mid-range temperature check.

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3. QUALIFICATION METHODOLOGY

Complete validation of the Kaye Validator AVS system will be demonstrated by the successful completion of both Installation and Operational Qualification Protocols. The Installation Qualification will be performed in the execution of a separate Validation Protocol. The Operational Qualification will be performed in the execution of this Validation Protocol and possible separate Operational Qualification protocols related to special software features.

This Operational Qualification is intended to verify and document that the Kaye Validator AVS system and its associated hardware and software components are fully operational. The Acceptance Criteria for each OQ procedure have been derived from the manufacturer's recommendations and/or cGMP requirements.

The purpose, methodology, and derivation of the Acceptance Criteria of each OQ procedure of this Validation Protocol are discussed below.

Note: The Kaye Validator AVS system is a portable instrument intended to be used throughout the facility on a routine basis. Although Amphenol Advanced Sensors specifies an operating environment of 0° to 50°C and a relative humidity of <95% (non-condensing), as well as a nominal supply voltage of 115 VAC or 230VAC, these environmental specifications will not be verified in this Installation Qualification because the instrument will not be permanently installed in a single location.

3.1. Software Security Definition (Section 6.1.)

The purpose of this OQ procedure is to specify and document the password options and test accounts that will be used in the execution of this Operational Qualification Protocol. 4 different Test accounts will be created and activated. This is to activate a user waiting for its initial login after being created by the System Administrator. Software forces newly created users to reset password after first login to make sure that the System Administrator does not know about the user's own password.

The test accounts will be deleted at the completion of OQ testing. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

3.2. Default System Administrator Account (Kaye/411) Deletion Verification / User List (Section 6.2.)

The purpose of this OQ procedure is to verify that the default System Administrator account (Kaye/411) has been deleted. The Kaye Validator AVS Software deletes this account automatically after the first login. This procedure will verify that the default System Administrator account may not log in using the default System Administrator (Kaye/411) account. Print a User List to check all active /disabled and deleted User Accounts. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

3.3. Equipment Communications Verification (Section 6.3.)

The purpose of this OQ procedure is to verify that the communication connections between the Kaye Validation Console, Kaye Validator AVS instrument, Kaye Validator AVS software and accessory equipment are functional. This will be accomplished by connecting the equipment and verifying that the correct serial numbers appear on the Hardware screen of the Kaye Validator AVS software. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

3.4. System to Instrument Data Transfer Verification (Section 6.4.)

The purpose of this OQ procedure is to verify that setups from the Kaye Validator AVS software can be successfully transferred to the Kaye Validator AVS instrument. Loading a setup directly from the software to the instrument will be tested. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

3.5. Instrument Inputs and Outputs Verification (Section 6.5.)

The purpose of this OQ procedure is to verify that all 48 inputs of the Kaye Validator AVS instrument are functional. At least one input will be placed in a Temperature Reference and monitored for an increase and decrease in temperature. The output relays will be activated over 100°C and verified with a Multimeter. The Acceptance Criteria for this section is derived from Amphenol Advanced Sensors recommendations.

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3.6. Basic Operating Function Verification (Section 6.6.)

The purpose of this OQ procedure is to verify that the basic functions of the Kaye Validator AVS system are operational. This will be accomplished by performing each of the five basic functions of the system (Setup programming, Calibration, Qualification study operation, Calibration Verification, and Report generation). The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

3.7. Instrument Loss of Power and Historical Data Testing Verification (Section 6.7.)

The purpose of this OQ procedure is to verify that the power backup function of the Kaye Validator AVS instrument performs as specified. This will be accomplished by disconnecting the AC power supply to the Kaye Validator AVS instrument while it is operating and verifying that the specified events occur as intended. The Historical Data is available after disconnecting and reconnecting the Kaye Validation Console to the Kaye Validator AVS instrument. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

3.8. Emergency Data Recovery via USB Side port (Section 6.8.)

The purpose of this OQ procedure is to verify that the Kaye Validator AVS instrument provides the possibility of Data Recovery via USB Side port. This will be accomplished by pressing the data transfer button without USB flash drive connected to the side port and by connecting a USB flash drive and transferring data to the flash drive by pressing data transfer button. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

3.9. File Management (Sync Out / Archive / Sync In / Access Study Files / AVS Convert) Functionality Verification (Section 6.9.)

The purpose of this OQ procedure is to verify that the Kaye Validator AVS system provides the possibility of Data Synchronization and Archiving and direct Upload of the last 10 study files from Kaye Validator AVS instrument to the Kaye Validation Console. This will be accomplished by performing and verifying the Sync Out, Archive and Sync In functions. In addition, data recovery of the last 10 studies using the Access Study File option and the ability to Convert the study files functionality will be verified by converting a study file. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

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4. RESPONSIBILITIES

Insert the names/titles/departments of all persons responsible for each of the following activities listed in Table 4.1.

Table 4.1. Responsibility Record

Activity	Person Responsible	Position/Title	Department
Protocol Execution			
Protocol Execution Review			
Discrepancy Review and Approval			
Report Generation			
Report Approval			

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5. EXECUTION REQUIREMENTS

Record all required data on the appropriate data sheets as it is collected. Record explanatory notes in the **Comments** section of the appropriate data sheets as necessary. Transcription errors may be corrected by crossing out the error with a single line, recording the corrected information, and initialing and dating the correction according to cGMP documentation requirements.

This Validation Protocol has been designed to verify and account for the purchased Kaye Validator AVS system as well as any other purchased or existing accessories such as IRTD, and Temperature References that will be used to execute this Protocol. In some cases, the OQ procedures refer to equipment, documentation, or other items beyond the scope of the system being validated. Verification items relating to optional Kaye Validator AVS system equipment and accessories that were not purchased or are not included with the Kaye Validator AVS system should be labeled "**Not Purchased**" or "**N/P**" to indicate that the equipment or accessory requires no documentation. Verification items that are extraneous or not relevant should be labeled "**Not Applicable**" or "**N/A**" to indicate that the specification is not relevant to the item being documented or verified. Verification items that are not currently available should be labeled "**Not Available**" or "**N/AV**".

Supporting data or documentation to be included with the executed Protocol and Final Report must be labeled with the Protocol number, a reference to the specific OQ procedure and/or Protocol page that the data supports, and the initials and date of the person who collected the supporting data.

If the supporting data or documentation is to be stored to CD or other media, the files will be named to identify the purpose of the file and/or the specific OQ procedure it supports. The CD will be labeled and have the initials and date of the person who collected and stored the supporting data.

Note: When generating reports, you may increase the Print Sampling rate before printing the report to reduce report size.

When a discrepancy is encountered during the execution of this protocol, complete a copy of the ***Validation Discrepancy Report Form*** according to the instructions in **Section 8**. Reference the discrepancy number in the ***Comments*** Section of the appropriate test suite. The discrepancy must be documented, resolved, and approved by the responsible reviewers before execution of this protocol can be considered complete and the final report can be approved.

Submit completed OQ procedures to the responsible reviewer on a timely basis.

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5.1. Signature Record

Each person involved in documenting, verifying, or approving any part of the execution of this Protocol (including data collection and Discrepancies) must record the specified information in Table 5.1.1 to positively identify all signatures and initials recorded in the execution of this Protocol. If a system for signature and initial identification is already in place, this signature record need not be completed.

Table 5.1.1. Signature Record

Printed Name	Signature	Initials	Position/Title	Date

5.2. Materials and Equipment

- Kaye Validator AVS instrument with Firmware Version 1.5
- At least one Kaye Validation Console with Software Version 1.5.1
- Minimum of one Kaye Sensor Input Module (SIM) with 12 Type T Thermocouples installed
- Kaye Temperature Reference
- Kaye IRTD
- Kaye Validator AVS User's Guide (Document #M5100)
- Multimeter

5.3. Training

Persons participating in the execution of this Validation Protocol must be adequately trained. Training requirements should include:

- Operation of the Kaye Validator AVS system and associated equipment
- Good Documentation Practices

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6. QUALIFICATION PROCEDURES

Complete each of the OQ procedures described on the following pages according to the specified procedures and all execution requirements listed in Section 5. Complete each data table and verify that the collected data conforms to the Acceptance Criteria for the corresponding OQ procedure. **NOTE:** All tests shall be executed with the user, **Test Supervisor**, unless instructed otherwise.

6.1. Software Security Definition

Objective:

The purpose of this OQ procedure is to specify and document the Administrator Preferences Settings, Administrator Policies Settings and test accounts that will be used during the execution of this Operational Qualification Protocol. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

Note: You must have a System Administrator account to perform this OQ procedure. When you start the program on the Kaye Validation Console for the first time, the Kaye Validator AVS software forces you to create a System Administrator account (see the *Kaye Validator AVS User's Guide or the Kaye Validator AVS Online Help* for more information). The default Admin at first Kaye Validator AVS Software start is Kaye/411.

Please create your own Admin account and use the account to login to the utility program in Section 6.1.1.

6.1.1. Administrator Preferences Settings

Procedure:

1. Start the **Kaye Validator AVS** Software.
2. **Login** with a valid System Administrator user ID and password. The **Activities** screen will appear.
3. On the **Activities** screen select **Admin** to access the **Administrator** site.
4. Select **Preferences** and specify the options as defined in Table 6.1.1.1. When finished, select **Save**.
5. Enter a valid Administrator user ID and password and select **OK** to confirm and save the changes.

Table 6.1.1.1. Administrator Preferences

Option	Specified	Actual is as Specified Y/N	Initials/Date
Company Name	_____		
Temperature	Celsius		
Max Groups	20		
Line Frequency	_____ Hz **Note**: Set to 60Hz in North America		
Pressure	Bar		
Allow Users to change Lethality	Yes		
IRTD Stability Threshold Lower Limit	400 °C		
Report Footer Setup/Cal/Verify	All Pages (checked)		

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6.1.2. Administrator Policies Settings

Procedure:

1. If you are still at the **Preferences** screen continue with Step 4 otherwise start the **Kaye Validator AVS** software.
2. **Login** with a valid System Administrator user ID and password. The **Activities** screen will appear.
3. On the **Activities** screen select **Admin** to access the **Administrator** site.
4. Select **Policies**.
5. Specify the options as defined in Table 6.1.2.1. When finished, select **Save**.
6. Enter a valid Administrator user ID and password and select **OK** to confirm and save the changes.

Table 6.1.2.1. Administrator Policies

Option	Specified	Actual is as Specified Y/N	Initials/Date
Require minimum length password	Enabled: 6 characters		
Expire passwords (days)	Enabled (180 days)		
Disable user account after 3 consecutive login failures	Enabled		
Display User ID during entry	Enabled		
Disable password system	Disabled		
Require Special Characters	Disabled		
Allow editing of D value in lethality calculation	Disabled		
Instrument calibration warning	Enabled for 12 Months		
Auto Sync In Users Before Sync Out	Disabled		

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6.1.3. User Account Definition

Procedure:

1. If you are still at the **Policies** screen continue with Step 4 otherwise start the Kaye Validator AVS software.
2. **Login** with a valid System Administrator user ID and password. The **Activities** screen will appear.
3. On the **Activities** screen select **Admin** to access the **Administrator** site.
4. Select **User Management** then **+ New User**.
5. Enter and verify the first user defined in Table 6.1.3.1. to include the specified User Privilege to **Copy Files/Reports** and **Change Console Time**. **Note:** User IDs and Passwords are case sensitive.
6. Repeat the process for each of the user accounts listed in Table 6.1.3.1. Verify that the entered users are accepted and listed on the **User Management** screen.
7. When completed, return to the **Activities** screen.

Table 6.1.3.1. User Account Definition

Name	User ID	Password	Title	User Type	Copy Files/ Reports	Change Console Time	Actual is as Specified Y/N	User Accepted and Listed Y/N	Initials/Date
Test System Administrator	**2222**	**111111**	Manager	System Administrator	Yes	No			
Test Supervisor	**3333**	**111111**	Attendant	Supervisor	Yes	Yes			
Test Operator 1	**4444**	**111111**	Executor	Operator	No	Yes			
Test Operator 2	**5555**	**111111**	Executor	Operator	No	Yes			

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6.1.4. User Account Activation

Procedure:

1. Open a login window (e.g. login to the Kaye Validator AVS or Reporting Software).
2. Enter the ID and password for the Test System Administrator account as specified in Table 6.1.4.1.
3. When prompted to change your password enter the new password as specified in Table 6.1.4.1.and record the results.
4. Repeat the process for each of the user account listed in table 6.1.4.1. and record the results.
5. When completed, return to the Activities screen.

Table 6.1.4.1. User Account Activation

Test Case	Codes Used	Specified Response	Actual is as Specified Y/N	Initials/Date
Test System Administrator Access	ID: **2222** Password: **111111**	First-time login - prompted to change password		
Test System Administrator Access: Change Password	New Password: **121212**	New password accepted		
Test Supervisor Access	ID: **3333** Password: **111111**	First-time login - prompted to change password		
Test Supervisor Access: Change Password	New Password: **121212**	New password accepted		
Test Operator 1 Access	ID: **4444** Password: **111111**	First-time login - prompted to change password		
Test Operator 1 Access: Change Password	New Password: **121212**	New password accepted		
Test Operator 2 Access	ID: **5555** Password:**111111**	First time login prompted to change password		
Test Operator 2 Access: Change Password	New Password:**121212**	New password accepted		

Comments:

_____

_____

_____

Software Security Definition (Section 6.1.)

Acceptance Criteria:

The Administrator Policies settings for the procedures in this Operational Qualification Protocol have been successfully defined. The Test Accounts that will be used during the OQ testing have been successfully created and activated. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

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6.2. Default System Administrator Account (Kaye/411) Deletion Verification / User List

Objective:

The purpose of this OQ procedure is to verify that the default System Administrator account may not log in using the default System Administrator (Kaye/411) account. Print a Users Report to check all Active/Disabled and Deleted User Accounts.

Procedure:

1. On the Kaye Validation Console, in the folder "Documents", create folder named "AVS IQ-OQ Supporting Documentation".

   Note: All documents supporting the execution of the IQ/OQ should be stored within the Folder unless otherwise specified in Section 7 of this Protocol.
2. Start the Kaye Validator AVS Software.
3. Attempt to log in to the system using the user ID **Kaye** and the password **411** as specified in Table 6.2.1. Record the results in Table 6.2.1.
4. Log in using the Test System Administrator as specified in Table 6.2.1. Record the results in Table 6.2.1.
5. Navigate to Administrator – User Management and check on the left pane that Default System Administrator Kaye is not listed.
6. Enter "Kaye" in the "Search" field and check that there is no search result. Record the results in Table 6.2.1.
7. Select **Test Operator 2** and delete the account.
8. Select **Users List** to generate the Users Report and verify that Test Operator 2 is listed as a deleted account. Record the results in Table 6.2.1.
9. Save the file as "**AVS OQ Section 6.2. – Users Report".**
10. When completed, return to the Activities screen.

Table 6.2.1. Administrator Level Access – Kaye Software

Test Case	Codes Used	Specified Response	Actual is as Specified Y/N	Initials/Date
Administrator Access System log in	ID: **Kaye** Password: **411**	Login unsuccessful, "Invalid Credential, Please try again" message appears.		
Administrator Access System log in	ID: **2222** Password: **121212**	Access allowed.		
Default System Administrator Kaye	N/A	**Kaye** is not listed in the available user list and no result when searched for **Kaye** account.		
Delete Test Operator 2	ID: 2222 Password: 121212	**Test Operator 2** not listed		
Print User List	N/A	**Test Operator 2** listed as deleted account		

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Comments:

_____

_____

_____

_____

Default System Administrator Account (Kaye/411) Deletion Verification / User List (Section 6.2.)

Acceptance Criteria:

The system will not allow users to log in using the default System Administrator account (Kaye/411). **Test Operator 2** is listed as a deleted account in the printed Users Report. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

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6.3. Equipment Communications Verification

Objective:

The purpose of this OQ procedure is to verify that the communication connections between the Kaye Validation Console, Kaye Validator AVS instrument, Kaye Validator AVS software and accessory equipment are functional. This will be accomplished by connecting the equipment and verifying that the correct serial numbers appear on the Hardware screen of the Kaye Validator AVS software.

Note: It is possible to connect a second IRTD if available and check communication. In that case the two IRTDs need to have different addresses! One IRTD Addressed 01 and the second IRTD Addressed 02. If no second IRTD is available or this option is not used mark the section in table below N/A for IRTD 2.

Important: It is recommended to connect or disconnect SIMs, IRTD(s) and or Temperature Reference while the Kaye Validator AVS is switched **off**!

6.3.1. USB/Docking Kaye Validation Console Communications

Procedure:

1. Connect up to 4 SIMs to the Kaye Validator AVS instrument.
   Note: In order to verify that all accessory equipment are functional, repeat this communication test if there are more than 4 SIMs.
2. Connect an IRTD and a Temperature Reference to the instrument. Record the serial numbers of all components in Table 6.3.1.1. where applicable.
3. Switch On the Kaye Validator AVS instrument. Wait until the Power LED on the front stops blinking which indicates that the booting process has finished. **Note:** This will take a few minutes.
4. Turn On the Kaye Validation Console, connect it directly to the Kaye Validator AVS instrument by docking it.
5. Login to the Kaye Validator AVS Software. Select **Discover** below Hardware.
6. Select **USB/Docking** and then select **Connect** at the bottom right of the screen. Verify that the top Communication LED on the front of the Kaye Validator AVS instrument is green, this will signalize communication between Kaye Validator AVS instrument and the Kaye Validation Console.
7. Select **Hardware Connection** on Live Data screen.
8. Communication is displayed with green lines in between the components. Select each component in order to display the detected Serial Number and record the displayed Serial Numbers of all components in Table 6.3.1.1.
   Note: For the Temperature Reference only the Model is displayed!
9. When completed, return to the Activities screen.

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Table 6.3.1.1. Equipment Communications

Communications Connection	Equipment Serial Number	Displayed Serial Number	Communication Functional Y/N	Initials/Date
Kaye Validator AVS instrument				
IRTD 1				
IRTD 2				
Temperature Reference Model: _____	N/A	N/A		
SIM # 1				
SIM # 2				
SIM # 3				
SIM # 4				
SIM # 5				
SIM # 6				
SIM # 7				
SIM # 8				

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6.3.2. WiFi Communications

Note: See User´s Guide M5100 to help configure the Kaye Validation Console and the Kaye Validator AVS to communicate via WiFi.

Procedure:

1. Switch On the Kaye Validator AVS instrument. Wait until the Power LED on the front stops blinking which indicates that the booting process has finished. **Note**: This will take a few minutes.
2. Turn On the Kaye Validation Console, Login to the Kaye Validator AVS Software. Select **Discover** below Hardware.
3. Connected to the Kaye Validator AVS instrument via WiFi. Select the appropriate AVS (blue button) and select **Connect**.
4. Once communication is established, **Live Data** screen is displayed and functional. Record the results in Table 6.3.2.1.
5. Verify that the top Communication LED on the front of the Kaye Validator AVS instrument is green, this signalizes communication between Kaye Validator AVS instrument and the Kaye Validation Console. Record the results in Table 6.3.2.1.
6. When completed, return to the Activities screen.

Table 6.3.2.1. WiFi Communications

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Connection via WiFi	Communication is established and **Live Data** screen is displayed and functional		
Kaye Validator AVS instrument signalizes communication	Communication LED on the front of the Kaye Validator AVS instrument turns on/green		

Comments:

_____

_____

_____

_____

Equipment Communications Verification (Section 6.3.)

Acceptance Criteria:

All equipment communication connections are functional. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

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6.4. System to Instrument Data Transfer Verification

Objective:

The purpose of this OQ procedure is to verify that Setups from the Kaye Validator AVS Software can be successfully transferred to the Kaye Validator AVS instrument.

Procedure:

1. Connect the Kaye Validation Console to the Kaye Validator AVS instrument and start the Kaye Validator AVS Software.
2. Using the Test Supervisor account, create a new Asset with the name "**OQ Tests**". Enter all necessary information (Type: Sterilizer) and Save the Asset.
3. Using the "OQ Tests" Asset define a Setup "**OQ Test Setup 1**" with 48 sensors. Under Sensors Configuration use temperature sensors which are available to connect during OQ (e.g. Type T TC (0.01)). Assign the Sensors a Label of "**Temp**" and select **Auto Numbering**.
4. Select "**Group Sensors**". Select Default Group to assign all sensors to one Group.
5. Select "**Calculations**". Define one Event: Message: Temp above 100; When: During entire cycle; Label: Hot TC; Condition: Max Value > 100. Select Add to save the Event.
6. Select "**Cal/Ver Parameters**". Keep the default Setpoints and enter the Stability criteria of Sensors for 3 Minutes, IRTD for 1 Minute.
7. Select "**Qualification Parameters**". Select "Hot TC" for Output Relay 1 and 2.
8. Select "**Review"** and save the Setup.
9. Select **Create Setup Report** and save it as PDF file in the "AVS IQ-OQ Supporting Documentation" folder with the file name "**AVS OQ Section 6.4. - Setup Report.**"
10. Return to the **Assets** screen and select the **OQ Tests** Asset.
11. Select "OQ Test Setup 1" and select **Initiate Qualification**. Select **OK** on the upcoming window.
12. Select **Connect** on the Select AVS screen. Confirm any upcoming message.
13. Check on the Qualification Study screen if the Setup Name "**OQ Test Setup 1"** is displayed. Record the result in Table 6.4.1.
14. When completed, return to the Activities screen

Table 6.4.1. System to Instrument Data Transfer

Kaye Validator AVS Setup	Specified Transfer Method	Current Setup Transferred to Kaye Validator AVS Y/N	Initials/Date
OQ Test Setup 1	Transfer by directly downloading from the Kaye Validation Console to the Kaye Validator AVS instrument		

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Comments:

_____

_____

_____

_____

System to Instrument Data Transfer Verification (Section 6.4.)

Acceptance Criteria:

System to instrument data transfer method is functional. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.5. Instrument Inputs and Outputs Verification

Objective:

The purpose of this OQ procedure is to verify the operation of the Output Relays and to verify that all 48 inputs of the Kaye Validator AVS instrument are functional.

This can be done by using a fully wired SIM inserted in all 4 slots consecutively or by using a fully wired SIM and 3 short-circuited Test SIMs. The Output Relays will be activated over 100°C and verified with a Multimeter. The functionality of the output relay will be tested by placing at least one Thermocouple into a Temperature Reference with a setpoint above 100°C.

Note: Power down the Kaye Validator AVS instrument before inserting a SIM into the SIM slot in order for the Kaye Validator AVS instrument to recognize the SIM. Also power down the Kaye Validator AVS instrument when disconnecting SIM.

Procedure:

1. Wire one SIM with 12 thermocouples.
2. With the Kaye Validator AVS instrument powered down, insert the fully wired SIM in slot #1 of the Kaye Validator AVS instrument and if available 3 short-circuited or wired SIMs in slot #2, #3 and #4 to test all the channels, and then power up the Kaye Validator AVS instrument.
   Note: If you have less than 4 SIMs, the SIMs available will be used to test slots #2, #3 and #4.
3. Connect the Multimeter leads to the Normally Open ("NO") and common output relays and set it to continuity or resistance mode.
4. Connect a Temperature Reference to the Kaye Validator AVS instrument.
5. With the Kaye Validation Console connected start the Kaye Validator AVS software (if not already running).
6. Load "OQ Test Setup 1" and start a qualification study when Temperature values are displayed. Select **Start Qual** and **Yes** and enter valid User Credentials. Confirm the “Maximum length of Study” message. (The output relays are only active while a qualification study is in progress).
7. Select **Hardware** and enter 120°C in the **Bath** temperature field and select **Set Temperature**. Record the results in Table 6.5.1.
8. Insert at least one thermocouple connected to the SIM in Slot 1 into the Temperature Reference, and observe the temperature reading response on the Kaye Validator AVS software display, and the response of the Multimeter monitoring the Output Relay. Verify the state of both Output Relays.
9. After the function of the Output Relays have been verified, remove the thermocouple from the Temperature Reference.
10. Again, verify the responses are as specified for the thermocouple and the Output Relays, record the results in Table 6.5.1.
11. Check if all channels are displayed and showing reasonable temperature values, record the results in Table 6.5.1.
12. Stop the qualification study once all inputs have been verified. Select **Stop Qual** and **Yes** and enter valid User Credentials. Select **Save Study** and enter valid User Credentials. Select **OK** on upcoming window. “Study File processed Successfully” message appears.
13. When completed, return to the Activities screen.

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When less than 4 SIMs are available

Procedure:

1. On the Activities screen, disconnect the hardware communications. Power off the Kaye Validator AVS, wait until red light stops blinking, transfer the available fully wired SIM(s) to the SIM slot(s) of the Kaye Validator AVS instrument that remain to be tested. Power On the Kaye Validator AVS instrument and re-establish hardware communications.
2. From the Live Data screen, check if all channels are displayed and showing reasonable temperature values, record the results in Table 6.5.1.
3. Repeat this procedure until all SIM slots have been tested.
4. When completed, return to the Activities screen.

Table 6.5.1. Instrument Inputs and Outputs

Input	Specified Response	Results are as Specified Y/N	Initials/Date
Temperature Reference	Set to 120°C		
Sensor Input Slot 1	Inserted Thermocouple's readings increase when inserted into the Temperature Reference and decrease when taken out of the Temperature Reference.		
Sensor Input Slot 1	Output relays active over 100°C and inactive when reading below 100°C (Both Output Relays are verified)		
Sensor Input Slot 1	All 12 Channel readings are displayed and showing reasonable temperature values		
Sensor Input Slot 2	All 12 Channel readings are displayed and showing reasonable temperature values		
Sensor Input Slot 3	All 12 Channel readings are displayed and showing reasonable temperature values		
Sensor Input Slot 4	All 12 Channel readings are displayed and showing reasonable temperature values		

Comments:

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Instrument Inputs and Outputs Verification (Section 6.5.)

Acceptance Criteria:

All instrument inputs and outputs function as specified. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.6. Basic Operating Function Verification

Objective:

The purpose of this OQ procedure is to verify that the basic functions of the Kaye Validator AVS system are operational. This will be accomplished by performing each of the five basic functions of the system (Setup programming, Calibration, Qualification study operation, Calibration Verification, and Report generation).

6.6.1. Setup Programming Verification

Procedure:

1. Login using the Test Supervisor and open the Asset “OQ Tests”.
2. Define a new setup named "**OQ Functional Test Setup**" with 12 Sensors. You may enter additional information in remaining fields for Setup programming verification.
3. Select “**Sensors Configuration**”. Select “Select Sensors” and select SIM 1 Sensor 1 to SIM 1 Sensor 12. Press ”**Select”**. First 12 Sensors are marked with an orange checkmark.
4. Configure Sensors with Sensor Type Thermocouple Type T TC (0.01). Sensor Label “TC” and check “**Auto Numbering**”. Select “**Assign**”.
5. Select “**Group Sensors**”. Select New Group and enter group name “All Sensors”.
6. Select all 12 TCs so that they all show an orange checkmark. Select Save.
7. Select “**Calculations**”. Check that by default all Statistical Calculations are selected.
8. Select “**Lethality Calculation**”. Define, Base Temperature 121.1°C, D-value 1.00, z-value 10 and Calculate Lethality “During Exposure Cycle”.
9. Select “**Interval Calculations**”. Define, **Interval Start = Exp Start** and **Interval End = Exp End** with all Calculations selected. Select **“Add**”.
10. Select “**Cal/Ver Parameters**”. Keep the default **Low**, **High** and **Check** Setpoints.
11. For the **Deviation Criteria,** set the **Calibrated Sensor** for 3 Minutes for both the Calibration Temperature Setpoints and Verification Temperature Setpoints.
12. For the **Stability Criteria**, set Sensors for 3 Minutes and IRTD for 1 Minute.
13. Select “**Qualification Parameters**”. Define, Start Qualification and Start Exposure “Manual”. Stop Exposure by Exposure Time(ExpTime) 0000:10:00 and Stop Qualification by Cycle Time 0000:30:00.
14. Set Data Storage, Sampling Rate: 10 seconds.
15. Set Clock Adjustment, Manual Adjustment: “At Qualification Start”
16. Set Relay, Output Relay1 and Output Relay2: “Unused”.
17. Select “**Review**” and select **Save**. Enter valid Supervisor credentials to save the Setup.
18. Select **Create Setup Report**. Enter valid Supervisor credentials and select OK.
19. Save the Setup report as a PDF in the “AVS IQ-OQ Supporting Documentation” folder with the file name “**AVS OQ Section 6.6.1. - Setup Report**”.
20. Open the PDF file “AVS OQ Section 6.6.1. - Setup Report” and verify each of the parameters listed in Table 6.6.1.1.
21. When completed, close the Report and return to the Activities screen.

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Table 6.6.1.1. Setup Report Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Header Section (First Page)	**Report Title:** Validator AVS Setup Report **Printed on:** dd-mmm-yyyy at hh:mm:ss by xxx **Study Name**: OQ Functional Test Setup **Company:** Not Specified - Information present **AVS SW Version:** Not specified – Information present **SOP Protocol #**: Information present in fields if entered during programming, otherwise blank **AVS Reports Version:** Not specified – Information present **Assets Name:** OQ Tests **Asset ID#:** Not Specified - Information present in fields if entered during programming, otherwise blank **Created by:** Test Supervisor **Date:** dd-mmm-yyyy hh:mm:ss (time zone) **Comments:** Not Specified - Information present in fields if entered during programming, otherwise blank **Last saved on:** dd-mmm-yyyy at hh:mm:ss by xxx		
Sensor Definitions	All 12 sensors programmed correctly SIM 2 to SIM 4 undefined		
Group Definition	Group Name: **All Sensors** **Asset ID#** Information present in fields if entered during programming, otherwise blank **SOP/Protocol Number**: Information present in fields if entered during programming, otherwise blank **Description**: Information present in fields if entered during programming, otherwise blank **Comment**: Information present in fields if entered during programming, otherwise blank Columns include: **SIM**, **Location** and **Sensor Name** 12 TCs listed as programmed		
Statistical Calculations	Statistical Calculations programmed correctly		
Lethality Calculations	Lethality Calculations programmed correctly		
Interval Calculations	Interval Calculations programmed correctly		
Calibration Parameters	Calibration parameters programmed correctly		

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Table 6.6.1.1. Setup Report Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Qualification Parameters	Qualification parameters programmed correctly		
Preferences	Not Specified - Information present		
Report Footer Section (every page)	**Performed by/Date** (unsigned) **Reviewed by/Date** (unsigned) Pages numbered: **Page x of x**		

Comments:

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Setup Programming Verification (Section 6.6.)

Acceptance Criteria:

Programming Setups with the Kaye Validator AVS software functions as specified. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.6.2. Calibration Process Verification

Procedure:

1. Connect an IRTD, Temperature Reference, and one SIM with 12 thermocouples (SIM slot #1) to the Kaye Validator AVS instrument. Record the serial numbers of all components in Table 6.6.2.1.
2. Insert all 12 thermocouples into the Temperature Reference and perform a calibration of all 12 thermocouples.
3. Open the Asset "OQ Tests" and under **Setups** select "OQ Functional Test Setup".
4. Select **Initiate Calibration**. Connect to the Kaye Validator AVS instrument.
5. Select the 12 TCs from SIM 1 and select **Initiate Calibration** again to proceed to the Sensor Calibration screen.
6. When Temperature values are displayed select **Start** and **Yes.** Type **OQ Calibration** in User Comment – Audit Trail field then enter valid User Credentials and select **OK**. (Record the Name of the User in Table 6.6.2.1). Confirm Warning message with **Yes**.
7. "Calibration Started Successfully" message will appear.
8. Record in Table 6.6.2.1 the "Calibration Started" time(hh:mm) displayed in the **Message** window.
9. Verify each of the Calibration Process parameters listed in Table 6.6.2.1.
10. When the calibration is complete select **Save Study**. Enter ID and password of user having started the Calibration. Select OK on the upcoming window. "Study File processed Successfully" message appears.
11. Return to the OQ Tests Asset and locate the "OQ Functional Test Setup" calibration study under **Calibrations**.
12. Generate a Calibration Report, save the Calibration report as a PDF in the "AVS IQ-OQ Supporting Documentation" folder with the file name "**AVS OQ Section 6.6.2. - Calibration Report**".
13. Open the PDF file "AVS OQ Section 6.6.2. – Calibration Report" and verify each of the parameters listed in Table 6.6.2.2.
14. When completed, close the Report and return to the Activities screen.

Table 6.6.2.1. Calibration Process Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Kaye Validator AVS instrument Serial Number	SN :_____		
SIM Serial Number	SN :_____		
IRTD Serial Number	SN :_____		
Temperature Reference Model	Model: _____		
User Comment – Audit Trail	OQ Calibration		
Access Security	ID and Password required to start Calibration		
Calibration User Name	Name: _____		
Calibration Start Time	Time: _____ : _____		

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Table 6.6.2.1 Calibration Process Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Temperature Reference Setpoint	Change to Low Setpoint 90°C		
Stabilization	Sensors and IRTD Stabilize at Low Setpoint. IRTD and Sensors Stability changes to green		
Sensor Calibration	Checklist on right side of display puts a green checkmark after each step is accomplished		
Graph view	Navigating to **Graph** view shows the positions of the Sensors to be calibrated against IRTD		
List view	Navigating to **List** view shows Setpoints and Checklist		
Setup	Navigating to **Setup** shows Setup Information		
Temperature Reference Setpoint	Change to High Setpoint 130°C		
Stabilization	Sensors and IRTD Stabilize at High Setpoint. IRTD and Sensors Stability changes to green		
Sensor Calibration	Checklist on right side of display puts a green checkmark after each step is accomplished		
Temperature Reference Setpoint	Change to Check Setpoint 121°C		
Stabilization	Sensors and IRTD Stabilize at Check Setpoint. IRTD and Sensors Stability changes to green		
Sensor Calibration	Checklist on right side of display puts a green checkmark after each step is accomplished		
Successful termination	Save Study button becomes enabled when calibration has been finished		
Study saved	“Study File processed Successfully” message appears		

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Table 6.6.2.2. Calibration Report Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Header Section (First Page)	**Report Title:** Validator AVS Sensor Calibration Report **Printed on:** dd-mmm-yyyy at hh:mm:ss by xxx **Study Name**: OQ Functional Test Setup **Company:** Not Specified - Information present **AVS SW Version:** Not specified – Information present **Asset Name:** OQ Tests **Asset ID #:** Not Specified - Information present if entered during programming, otherwise blank **AVS Reports Version:** Not specified – Information present **Started by:** Same as recorded in Table 6.6.2.1. **Sensor Calibration Date:** Not Specified - Information present **Firmware Version:** Not specified – Information present **Setup Created By:** Test Supervisor **Setup Created Date:** dd-mmm-yyyy hh:mm:ss (time zone) **DAQ Version:** Not specified – Information present **AVS Validator S/N:** Same as recorded in Table 6.6.2.1. **Comments:** Not Specified - Information present if entered during programming, otherwise blank		
Page 1 (Cover Page)	**Sensor Input modules (SIMs)**: Same as recorded in Calibration Process Verification **Calibration setpoints**: Low 90.0°C, High 130.0°C, Check 121.0°C **Stability criteria**: Sensor 0.20°C for 3 min, IRTD 0.012°C for 1 min **Deviation criteria**: Uncalibrated Sensor 1.00°C, Calibrated Sensor 0.50°C, Log corrected results 3 minutes **Sensor calibration initiated on**: Shown as dd-mmm-yyyy hh:mm:ss **Calibration User Name:** Same as recorded in Table 6.6.2.1. IRTD **Serial Number** and **Calibration Date**: Same S/N as recorded in Table 6.6.2.1. **Temperature reference**: Same Model as recorded in Table 6.6.2.1.		

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Table 6.6.2.2. Calibration Report Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Low Calibration Point **90.0°C**	Stability Evaluation Data: Not Specified – Information present Low Calibration Point Stabilization Time: ≥ 3:00 minutes at ~90.0°C Deviation Evaluation Data: Not Specified – Information present Low Calibration Point - Corrected Results are logged: 3:00 minutes at ~90.0°C Low Calibration Point Sensor Deviation: ≤ 0.50°C		
High Calibration Point **130.0°C**	Stability Evaluation Data: Not Specified – Information present High Calibration Point Stabilization Time: ≥ 3:00 minutes at ~130.0°C Deviation Evaluation Data: Not Specified – Information present High Calibration Point - Corrected Results are logged: 3:00 minutes at ~130.0°C High Calibration Point Sensor Deviation: ≤ 0.50°C		
Check Calibration Point **121.0°C**	Stability Evaluation Data: Not Specified – Information present Check Calibration Point Stabilization Time: ≥ 3:00 minutes at ~121.0°C Deviation Evaluation Data: Not Specified – Information present Check Calibration Point - Corrected Results are logged: 3:00 minutes at ~121.0°C Check Calibration Point Sensor Deviation: ≤ 0.50°C		
Report Footer Section (every page)	**Performed by/Date** (unsigned) **Reviewed by/Date** (unsigned) Pages numbered: **Page x of x**		

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Comments:

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Calibration Process Verification (Section 6.6.2.)

Acceptance Criteria:

The Calibration process and Calibration reporting function as specified. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.6.3. Qualification Process Verification

Procedure:

1. Make sure the Temperature Reference Setpoint is at 90°C.
2. With the Kaye Validation Console connected to the Kaye Validator AVS, navigate to the Asset OQ Tests and Setups.
3. Start a qualification study using "OQ Functional Test Setup." Select Setup **OQ Functional Test Setup** and select **Initiate Qualification.** Select **OK** on upcoming window.
4. Connect to the Kaye Validator AVS on the Select AVS screen. Select **OK** to "Accept current Console time", verify the qualification process parameter listed in Table 6.6.3.1.
5. Select **Start Qual/Yes** on Qualification study screen when Temperature values are displayed below "Value." Type **OQ Qualification** in User Comment – Audit Trail field, select the "**Only I can stop this study**" checkbox then enter valid User Credentials and select **OK**. Record the Name of the User in Table 6.6.3.1.
6. Confirm "Maximum length of Study" message, select **OK**.
7. Record the Start Time in Table 6.6.3.1. from the "Manual Qual Started" time displayed in the Message window on the Kaye Validation Console display.
8. Change the setpoint of Temperature Reference to 122.5°C.
9. Verify each of the qualification process parameters listed in Table 6.6.3.1.
10. Within 10 minutes of the qualification start time, start the Exposure cycle (once the thermocouple readings are all at least above 100°C). Record the time in Table 6.6.3.1.
11. When Exposure ends automatically after 10 minutes set the Temperature Reference to **95°C** Setpoint.
12. Select **Quit Qual/Yes** and enter the User Credentials of a valid Operator account that was not used to start the Qualification study. Record the Name of the User in Table 6.6.3.1.
13. Verify that the stopping of the Study is not permitted by a User that did not start the Qualification study. Document the results in Table 6.6.3.1.
14. Verify each of the remaining qualification process parameters listed in Table 6.6.3.1.
15. When the qualification study is complete, select **Save Study**, enter User ID and Password. Select **OK** on upcoming window. "Study File processed Successfully" message appears.
16. Return to the OQ Tests Asset and locate the "OQ Functional Test Setup" Qualification study under **Qualifications**.
17. Generate a Qualification Detailed Report, Qualification Summary Report and Interval Calculations Report.
18. Accept the defaults when the "Mark Qualification Start and End" and the "Marking Exposure" screens appear. Select **Next** on Cycle Time window and select **No** on "Edit Cycle Name Alert" window.
19. Save the Reports as a PDF in the "AVS IQ-OQ Supporting Documentation" folder with the file names "**AVS OQ Section 6.6.3. – Detailed Report**", "**AVS OQ Section 6.6.3. – Summary Report**" and "**AVS OQ Section 6.6.3. – Interval Calcs Report**".
20. Verify the report parameters listed in Tables 6.6.3.2., 6.6.3.3 and 6.6.3.4.
21. When completed, close the Report and return to the Activities screen.

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Table 6.6.3.1. Qualification Process Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
"Set AVS System Clock and Console Time" window appears	Accept current Console time		
User Comment – Audit Trail	OQ Qualification		
Access Security	ID and Password required to start Qualification		
Qualification Study Start Time	Time: _____ : _____		
Qualification User Name	Name: _____		
Status LED on front of AVS	Status LED green after starting study		
Graph view	Navigating to **Graph** view shows Sensors' Trend		
Graph view	Select **Manual Scale** to set Max and Min Scale – select **Reset**		
Graph view	Select **Limit** to set High and Low Limit Lines – select **Reset**		
Graph view	Use your fingers to **Zoom** into the Graph		
List view	Return to **List** view to bring up Values, Messages and Calculations		
Setup	Navigating to **Setup** shows Setup Information		
Manual Exposure Start	Within 10 minutes of study start Time: _____ : _____ : _____		
Accumulated Lethality	Calculation starts for each sensor during exposure cycle		
Graph view	Select Lethality and select LineSeries or BarSeries to show Lethality (Calculation starts for each sensor during exposure cycle).		
Interval Calculation	Shows Sensor data, Interval Start, Interval End and Interval Duration		
Automatic Exposure End	10:00 minutes after Exposure start		
User Name used in the Quit Qual attempt	Name: _____		
Attempt to Quit Qual	"User who started the qualification can only stop the qualification" message appears.		
Statistical Calculations	Displayed and updated throughout cycle		

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Table 6.6.3.1. Qualification Process Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Study Min&Max	Shows Min/Max data for the entire Study		
Automatic Qualification End	30 minutes after qualification study start		
Status LED on front of AVS	Status LED off after study is stopped		
Successful termination	Save Study button becomes enabled when qualification has been finished		
Saving Study	"Study File processed Successfully" message appears		

Table 6.6.3.2. Qualification Detailed Report Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Header Section (First Page)	**Report Title:** Validator AVS Qualification Detailed Report **Printed on:** dd-mmm-yyyy at hh:mm:ss by xxx **Study Name**: OQ Functional Test Setup **Company:** Not Specified - Information present **AVS SW Version:** Not specified – Information present **SOP / Protocol #:** Not Specified - Information present if entered during programming, otherwise blank **Run #**: Not Specified - Information present **AVS Reports Version:** Not specified – Information present **Asset Name:** OQ Tests **Asset ID #:** Not Specified - Information present if entered during programming, otherwise blank **Firmware Version:** Not specified – Information present **Started by:** Same as recorded in Table 6.6.3.1. **Date:** dd-mmm-yyyy hh:mm:ss (time zone) **DAQ Version:** Not specified – Information present **Setup Created By:** Test Supervisor **Setup Created Date:** dd-mmm-yyyy hh:mm:ss (time zone) **AVS Validator S/N:** Same as recorded in Table 6.6.2.1. **Comments:** Not Specified - Information present if entered during programming, otherwise blank		
Report Header Section (all other pages)	**Report Title:** Validator AVS Qualification Detailed Report **Printed on**: dd-mmm-yyyy at hh:mm:ss by xxx **Study Name:** OQ Functional Test Setup **SOP/Protocol #:** Not Specified - Information present if entered during programming, otherwise blank		

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Table 6.6.3.2. Qualification Detailed Report Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Footer Section (every page)	**Performed by/Date** (unsigned) **Reviewed by/Date** (unsigned) Pages numbered: **Page x of x**		
Page 1 or 2 (Cover Page)	Contains "**Comments:**" section, and "**Sensor Input Modules (SIMs)**" section with AVS Validator and SIM serial numbers		
Page 1 or 2	Contains "**Groups Included in Report**", and "**Sensors included in Report**" section, listing all Sensors and Groups. Columns include Group Type, Sensor S/N, Probe ID, FW Version, Sensor Description, Calculate Lethality and Battery Life.		
Report Header Section (all other pages)	Group Name: **All Sensors** **Asset ID:** Not Specified - Information present if entered during programming, otherwise blank **SOP Protocol:** Not Specified – Information present if entered during programming, otherwise blank **Comments:** Not Specified - Information present if entered during programming, otherwise blank **Description:** Not Specified - Information present if entered during programming, otherwise blank		
Temperature Data (for each group)	Contains Labels of each Sensor in the test that contains a Temperature Sensor and unit of measurement, across the page, with the raw data listed in column fashion below.		
Temperature Data (for each group)	Verify that for each Sensor, the samples are time stamped by a consistent number of seconds or minutes based on the selected Print Sampling Rate and includes any applicable off Print Sampling Rate Events.		
Qualification **Study Start** Time	Same as recorded in Qualification Process Verification		
Start Exposure Event	Same as recorded in Qualification Process Verification		
Automatic **End Exposure** Event	10:00 minutes after Exposure start		
Automatic **Study End** Event	30:00 minutes after Qualification Run start		
Temperature Data(ºC) - Statistical Calculations	Contains the following calculations: **Min, S/N Min, Max, S/N Max, Max-Min, Avg, Std Dev, Max-Avg, Avg-Min** and **Cycle Time**. Calculated throughout cycle Also contains: **Study Start, Start Exposure, End Exposure,** and **Study End** events.		

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Table 6.6.3.2. Qualification Detailed Report Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Lethality Data	Contains the Lethality data for each sensor and the following calculations: **Min, S/N Min, Max, S/N Max** and **Cycle Time** Calculated during Exposure cycle Also contains: **Study Start, Start Exposure, End Exposure,** and **Study End** events.		
Last page	**Comments:** Not Specified - Information present if entered during programming, otherwise blank **System Messages**: 'Study Start', 'Start Exposure', 'End Exposure', and 'Study End' with corresponding Console ID.		

Table 6.6.3.3. Qualification Summary Report Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Header Section (First Page)	**Report Title:** Validator AVS Qualification Summary Report **Printed on:** dd-mmm-yyyy at hh:mm:ss by xxx **Study Name**: OQ Functional Test Setup **Company:** Not Specified - Information present **AVS SW Version:** Not specified – Information present **SOP / Protocol #:** Not Specified - Information present if entered during programming, otherwise blank **Run #:** Not Specified - Information present **AVS Reports Version:** Not specified – Information present **Asset Name:** OQ Tests **Asset ID #:** Not Specified - Information present if entered during programming, otherwise blank **Firmware Version:** Not specified – Information present **Started by:** Same as recorded in Table 6.6.3.1. **Date:** dd-mmm-yyyy hh:mm:ss (time zone) **Setup Created By:** Test Supervisor **DAQ Version:** Not specified – Information present **Setup Created Date:** dd-mmm-yyyy hh:mm:ss (time zone) **AVS Validator S/N:** Same as recorded in Table 6.6.2.1. **Comments:** Not Specified – Information present if entered during programming, otherwise blank		
Report Header Section (all other pages)	**Report Title:** Validator AVS Qualification Summary Report **Printed on**: dd-mmm-yyyy at hh:mm:ss by xxx **Study Name:** OQ Functional Test Setup **SOP/Protocol #:** Not Specified - Information present if entered during programming, otherwise blank		

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Table 6.6.3.3. Qualification Summary Report Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Footer Section (every page)	**Performed by/Date** (unsigned) **Reviewed by/Date** (unsigned) Pages numbered: **Page x of x**		
Page 1 or 2 (Cover Page)	Contains "**Comments:**" section, and "**Sensor Input Modules (SIMs)**" section with AVS Validator and SIM serial numbers		
Page 1 or 2	Contains "**Groups Included in Report**", and "**Sensors included in Report**" section, listing all Sensors and Groups. Columns include Group Type, Sensor S/N, Probe ID, FW Version, Sensor Description, Calculate Lethality and Battery Life.		
Group Header (Beginning on Page 3)	Group Name: **All Sensors** **Asset ID:** Not Specified - Information present if entered during programming, otherwise blank **SOP Protocol:** Not Specified – Information present if entered during programming, otherwise blank **Comments:** Not Specified - Information present if entered during programming, otherwise blank **Description:** Not Specified - Information present if entered during programming, otherwise blank		
"**Temperature Data(ºC)**" section (Beginning on Page 3)	Sensor/Logger SN used in the study with the following calculations**:** **Min, Max, Avg, Cycle ALeth** (if defined for the group) and **Max-Min** for each Temperature Sensor in the study and calculated throughout cycle.		
"**Temperature Summary Data(ºC)**" section (Beginning on Page 3)	Contains values for: **Cycle Start** **Cycle Duration** **Min of Min** (with S/N: label) **Time** (for Min of Min) **Max of Max** (with S/N: label) **Time** (for Max of Max) **Max Range** **Max Spread/Time** **Min ALeth** (with S/N: label) **Max ALeth** (with S/N: label) **Avg of Avg** **Max(Max-Min)** (with S/N: label)		
"**Totals**" section (2nd to Last Page)	**Total ALeth** (Total calculated lethality for each sensor**)**		

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Table 6.6.3.3. Qualification Summary Report Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
"**Totals**" section (2nd to Last Page)	Contains values for: **Study Start** **Study End** **Duration** **Min ALeth** **S/N (**Min ALeth Sensor label**)** **Max ALeth** **S/N** (Max ALeth Sensor label**)** **Study Min** **S/N** (Study Min Sensor label) **Study Max** **S/N** (Study Max Sensor label)		
Last page	**Comments:** Information present if entered during programming, otherwise blank **System Messages:** 'Study Start', 'Start Exposure', 'End Exposure', and 'Study End'.		

Table 6.6.3.4. Interval Calculations Report Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Header Section (First Page)	**Report Title:** Validator AVS Interval Calculations Report **Printed on:** dd-mmm-yyyy at hh:mm:ss by xxx **Study Name**: OQ Functional Test Setup **Company:** Not Specified - Information present **AVS SW Version:** Not specified – Information present **SOP / Protocol #:** Not Specified - Information present if entered during programming, otherwise blank **Run #:** Not Specified - Information present **AVS Reports Version:** Not specified – Information present **Asset Name:** OQ Tests **Asset ID #:** Not Specified - Information present if entered during programming, otherwise blank **Firmware Version:** Not specified – Information present **Started by:** Same as recorded in Table 6.6.3.1. **Date:** dd-mmm-yyyy hh:mm:ss (time zone) **DAQ Version:** Not specified – Information present **Setup Created By:** Test Supervisor **Setup Created Date:** dd-mmm-yyyy hh:mm:ss (time zone) **AVS Validator S/N:** Same as recorded in Table 6.6.2.1. **Comments:** Not Specified – Information present if entered during programming, otherwise blank		

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Table 6.6.3.4. Interval Calculations Report Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Header Section (all other pages)	**Report Title:** Validator AVS Interval Calculation Report **Printed on**: dd-mmm-yyyy at hh:mm:ss by xxx **Study Name:** OQ Functional Test Setup **SOP/Protocol #:** Not Specified - Information present if entered during programming, otherwise blank		
Report Footer Section (every page)	**Performed by/Date** (unsigned) **Reviewed by/Date** (unsigned) Pages numbered: **Page x of x**		
Page 1 or 2 (Cover Page)	Contains "**Comments:**" section, and "**Sensor Input Modules (SIMs)**" section with AVS Validator and SIM serial numbers		
Page 1 or 2	Contains "**Groups Included in Report**", and "**Sensors included in Report**" section, listing all Sensors and Groups. Columns include Group Type, Sensor S/N, Probe ID, FW Version, Sensor Description, Calculate Lethality and Battery Life.		
Group Header (Beginning on Page 2)	Group Name: **All Sensors** **Asset ID:** Not Specified - Information present if entered during programming, otherwise blank **SOP Protocol:** Not Specified – Information present if entered during programming, otherwise blank **Comments:** Not Specified - Information present if entered during programming, otherwise blank **Description:** Not Specified - Information present if entered during programming, otherwise blank		
Interval Calculations information	Interval parameters to include: **Interval Start, Interval End and Interval Duration.**		
Interval Calculations Data	Contains the following information and calculations: **Sensors**, **Min(ºC)**, **Min Time**, **Max(ºC)**, **Max Time** and **Avg(ºC)**.		
Interval Calculations Summary Data	Contains the following information and calculations: **Min of Min** and **Max of Max.**		
Last page	**Comments:** Information present if entered during programming, otherwise blank **System Messages:** 'Study Start', 'Start Exposure', 'End Exposure', and 'Study End' with corresponding Console ID.		

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Comments:

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Qualification Process Verification (Section 6.6.3.)

Acceptance Criteria:

The Qualification process and Qualification reporting function as specified. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.6.4. Calibration Verification Process Verification

Procedure:

1. Connect the Temperature Reference to the Kaye Validator AVS instrument and insert all 12 thermocouples into the Temperature Reference in preparation to perform a Calibration Verification of all 12 thermocouples.
2. Navigate to the Asset "OQ Tests" and under "Setups", select "OQ Functional Test Setup."
3. Select **Initiate Verification**. Connect to the Kaye Validator AVS instrument.
4. Select the 12 TCs from SIM 1 and select **Initiate Verification** again to advance to the Sensor Verification screen.
5. Select **Setup** and change the Low Setpoint to 95°C, the High Setpoint to 129°C and the Check Point to 121.5°C. Set the Deviation Criteria for Calibrated Sensors to 0.6°C. Select **OK** and enter valid User Credentials. Select **OK**, "Data saved successfully" message will appear.
6. When Temperature values are displayed select **Start** and **Yes.** Type **OQ Verification** in User Comment – Audit Trail field then enter valid User Credentials and select **OK**. Record the Name of the User in Table 6.6.4.1.
7. "Sending Setup to AVS" message will appear, followed by "Verification Started Successfully" message.
8. Record in Table 6.6.4.1. the "Verification Started" time(hh:mm) displayed in the **Message** window.
9. Verify each of the Calibration Verification process parameters listed in Table 6.6.4.1.
10. When the Calibration Verification is complete select **Save Study**. Enter ID and Password of user having started the Calibration Verification. Select **OK** on the upcoming window. "Study File processed Successfully" message appears.
11. Return to the OQ Tests Asset and locate the "OQ Functional Test Setup" calibration verification study under **Verifications**.
12. Generate a Sensor Verification Report, save the Sensor Verification report as a PDF in the "AVS IQ-OQ Supporting Documentation" folder with the file name "**AVS OQ Section 6.6.4. – Sensor Verification Report**".
13. Open the PDF file "AVS OQ Section 6.6.4. – Sensor Verification Report" and verify each of the parameters listed in Table 6.6.4.2.
14. When completed, close the Report and return to the Activities screen.

Table 6.6.4.1. Calibration Verification Process Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Kaye Validator AVS instrument Serial Number	Same as recorded in Calibration Process Verification		
SIM Serial Number	Same as recorded in Calibration Process Verification		
IRTD Serial Number	Same as recorded in Calibration Process Verification		
Temperature Reference Model	Same as recorded in Calibration Process Verification		
User Comment – Audit Trail	OQ Verification		
Access Security	ID and Password required to start Calibration		
Calibration Verification User Name	Name: _____		
Calibration Verification Start Time	Time: _____ : _____		

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Table 6.6.4.1. Calibration Verification Process Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Temperature Reference Setpoint	Changes to Low Setpoint 95°C		
Stabilization	Sensors and IRTD Stabilize at Low Setpoint. IRTD and Sensors Stability changes to green		
Sensor Calibration Verification	Checklist on right side of display puts a green checkmark after each step is accomplished		
Graph view	Navigating to **Graph** view shows the positions of the Sensors to be calibrated against IRTD		
List view	Navigating to **List** view shows Setpoints and Checklist		
Setup	Navigating to **Setup** shows the verification parameters		
Temperature Reference Setpoint	Changes to High Setpoint 129°C		
Stabilization	Sensors and IRTD Stabilize at High Setpoint. IRTD and Sensors Stability changes to green		
Sensor Calibration Verification	Checklist on right side of display puts a green checkmark after each step is accomplished		
Temperature Reference Setpoint	Changes to Check Setpoint 121.5°C		
Stabilization Verification	Sensors and IRTD Stabilize at Check Setpoint. IRTD and Sensors Stability changes to green		
Sensor Calibration Verification	Checklist on right side of display puts a green checkmark after each step is accomplished		
Successful termination	Save Study button becomes enabled when calibration verification has been finished		
Saving Study	“Study File processed Successfully” message appears		

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Table 6.6.4.2. Calibration Verification Report Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Report Header Section (First Page)	**Report Title:** Validator AVS Sensor Verification Report **Printed on:** dd-mmm-yyyy at hh:mm:ss by xxx **Study Name**: OQ Functional Test Setup **Company:** Not Specified - Information present **AVS SW Version:** Not specified – Information present **Assets Name:** OQ Tests **Asset ID #:** Not Specified - Information present if entered during programming, otherwise blank **AVS Reports Version:** Not specified – Information present **Started by:** Same as recorded in Table 6.6.4.1. **Sensor Verification Date:** Not Specified - Information present **Firmware Version:** Not specified – Information present **Setup Created By:** Test Supervisor **Setup Created Date:** dd-mmm-yyyy hh:mm:ss (time zone) **DAQ Version:** Not specified – Information present **AVS Validator S/N:** Same as recorded in Table 6.6.2.1. **Comments:** Not Specified - Information present if entered during programming, otherwise blank		
Page 1 (Cover Page)	**Sensor Input modules (SIMs)**: Same as recorded in Table 6.6.2.1. **Verification setpoints**: Low 95.0°C, High 129.0°C, Check 121.5°C **Stability criteria**: Sensor 0.20°C for 3 min, IRTD 0.012°C for 1 min **Deviation criteria:** Calibrated Sensor 0.60°C, Log corrected results 3 min **Calibration verification initiated on:** Shown as dd-mmm-yyyy hh:mm:ss **Verification User Name:** Same as recorded in Table 6.6.4.1. IRTD **Serial Number** and **Calibration Date**: Same S/N as recorded in Table 6.6.2.1. **Temperature reference**: Same Model as recorded in Table 6.6.2.1.		

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Table 6.6.4.2. Calibration Verification Report Verification (continued)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Low Verification Point **95.0°C**	Stability Evaluation Data: Not Specified – Information present Low Verification Point Stabilization Time: ≥ 3:00 minutes at ~95.0°C Deviation Evaluation Data: Not Specified – Information present Low Verification Point - results are logged: 3:00 minutes at ~95.0°C Low Verification Point Sensor Deviation: ≤ 0.60°C		
High Verification Point **129.0°C**	Stability Evaluation Data: Not Specified – Information present High Verification Point Stabilization Time: ≥ 3:00 minutes at ~129.0°C Deviation Evaluation Data: Not Specified – Information present High Verification Point - results are logged: 3:00 minutes at ~129.0°C High Verification Point Sensor Deviation: ≤ 0.60°C		
Check Verification Point **121.5°C**	Stability Evaluation Data: Not Specified – Information present Check Verification Point Stabilization Time: ≥ 3:00 minutes at ~121.5°C Deviation Evaluation Data: Not Specified – Information present Check Verification Point - results are logged: 3:00 minutes at ~121.5°C Check Verification Point Sensor Deviation: ≤ 0.60°C		
Report Footer Section (every page)	**Performed by/Date** (unsigned) **Reviewed by/Date** (unsigned) Pages numbered: **Page x of x**		

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Comments:

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Calibration Verification Process Verification (Section 6.6.4.)

Acceptance Criteria:

The Calibration Verification process and Calibration Verification reporting function as specified. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.6.5. Kaye Validation Console controlled Operation Audit Trail Verification

Procedure:

1. On the Kaye Validator AVS **Activities** screen select on **Audit**.
2. Open Date Filter "Is greater than or equal to" the day you performed the Calibration, and "Is less than or equal to" the day you performed the Calibration Verification. Select **Filter**.
3. Generate a Report of the filtered Audit Trail. Save the file as a PDF file in the, "AVS IQ-OQ Supporting Documentation" folder with the file name, "**AVS OQ Section 6.6.5. – Audit Trail Report**".
4. Verify that the specified results in Table 6.6.5.1 are present in the Audit Trail Report.
5. When completed, close the Report and return to the Activities screen.

Table 6.6.5.1. Audit Trail Verification

Test Case	Specified Result	Actual is as Specified Y/N	Initials/Date
Calibration Study	The loading of OQ Functional Test Setup and the start of the Calibration is captured in the Audit Trail with the User Name, User ID, date and time. **OQ Calibration** in User Comment - Audit Trail.		
Save Study	Calibration file Saved for OQ Functional Test Setup with the User Name, User ID, date and time.		
Qualification Study	The loading of OQ Functional Test Setup, the start of the Qualification, the start of Exposure and the automatic termination of the Qualification is captured in the Audit Trail with the User Name, User ID, date and time. **OQ Qualification** in User Comment - Audit Trail.		
Save Study	Qualification file Saved for OQ Functional Test Setup with the User Name, User ID, date and time.		
Calibration Verification Study	The loading of OQ Functional Test Setup, its modification and the start of the Calibration Verification is captured in the Audit Trail with the User Name, User ID, date and time. **OQ Verification** in User Comment - Audit Trail.		
Save Study	Calibration Verification file Saved for OQ Functional Test Setup with the User Name, User ID, date and time.		

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Comments:

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Kaye Validation Console controlled Operation Audit Trail Verification (Section 6.6.5.)

Acceptance Criteria:

The Calibration, Qualification and Calibration Verification processes and reporting are recorded in the Audit Trail. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.7. Instrument Loss of Power and Historical Data Testing Verification

Objective:

This test will verify that the Kaye Validator AVS instrument backup battery operates as specified. The backup battery provides short-term power in case of AC power loss during a Qualification study. The Historical Data availability will be tested by disconnect and reconnect the Kaye Validation Console after a view Minutes. Record all times according to the clock on the Kaye Validation Console (where possible).

Procedure:

1. Login to the Kaye Validator AVS Software with the Test Supervisor account. Open/**Edit** the Setup "OQ Functional Test Setup" and modify the Setup.
2. Under Calculations, change the Lethality Calculation to **Undefined** and delete Interval Calculations.
3. Under Qualification Parameters, change the Start Exposure and Stop Exposure to **Undefined**, change the Stop Qualification to **Manual** and change the Clock Adjustment/Manual Adjustment to **Disabled**.
4. Under Review, select Copy as New Setup, save as "**OQ Functional Test Setup-Power Loss**" then return to the Asset.
   Note: When exiting back to the Asset select **Yes** when the "You are about to lose your changes. Do you want to continue?" message appears.
5. Start a qualification study using "OQ Functional Test Setup-Power Loss". Record the Start time in Table 6.7.1.
6. During the study, disconnect the power supply to the instrument. Record the time in Table 6.7.1.
   Note: To disconnect power to the instrument, Amphenol Advanced Sensors recommends plugging the instrument into a power strip and turning off the power strip.
7. Verify the sequence of Loss of Power events and document the events in Table 6.7.1.
8. **Disconnect** the Kaye Validation Console from AVS for at least 5 minutes. – Communication LED turns off.
9. **Reconnect** the Kaye Validation Console to the AVS – Communication LED turns on.
10. Check availability of Historical Data on **Graph** view.
11. After at least another 10 minutes reconnect the power. Record the time and verify the sequence of events in Table 6.7.1.
12. Stop the Qualification, record the Time, verify and document the events in Table 6.7.1.
13. Save the Study, verify and document the events in Table 6.7.1.
14. Generate a Qualification Detailed Report. Use default Start and End.
15. Save the file as a PDF file in the, "AVS IQ-OQ Supporting Documentation" folder with the file name, "**AVS OQ Section 6.7. – Detailed Report**".
16. View the Report and verify that all Data from Qual Start to End is available. Verify "AVS On Battery" and "AVS On AC" messages are present in the Report. Document the results in Table 6.7.1
17. When completed, close the Report and return to the Activities screen.

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Table 6.7.1. Loss of Power and Historical Data

Verification Parameters	Results are as Specified Y/N	Initials/Date
Manual Qual Started: _____ : _____ : _____		
Power Disconnected: _____ : _____		
Power LED on AVS Front panel switches from green to red		
The Kaye Validator AVS produces an audible beep indicating power loss		
Kaye Validator AVS continuously collects data		
AVS on Battery message: _____ : _____ : _____		
Disconnect Kaye Validation Console from AVS		
Communication LED turns off		
Select **Hardware** – Discover AVS – USB Docking – Connect to AVS		
Communication LED turns on		
Graph view - Wait until the scale for Historical Data appeared		
Select the left arrow to load Historical Data		
Data is available from the Start of Qualification		
Power Reconnected: _____ : _____		
Power LED on AVS Front panel switches from red to green		
AVS on AC message: _____ : _____ : _____		
Manual Qual Stopped: _____ : _____ : _____		
Study can be saved after power has been reconnected		
AVS File Saved: _____ : _____ : _____		
Qualification report can be generated, all data is available with and “AVS On Battery” and “AVS On AC” messages are present in the Report		

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Comments:

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Instrument Loss of Power and Historical Data Testing Verification (Section 6.7.)

Acceptance Criteria:

The Kaye Validator AVS instrument backup battery provides short-term power in case of AC power loss during a Qualification study. The Kaye Validator AVS instrument continuously collects data when the AC power failed. The Historical Data is available after disconnecting and reconnecting the Kaye Validation Console to the Kaye Validator AVS instrument. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

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6.8. Emergency Data Recovery via USB Side Port Functionality Verification

Objective:

This test will verify that the Kaye Validator AVS instrument provides the possibility of Data Recovery via USB Side port. On its left side, the Kaye Validator AVS instrument offers a side port with a USB port to accommodate a USB flash drive for Data Recovery and a button to initiate the data transfer.

Note: Make sure that at least one or more Calibration, Qualification or Verification study have been performed with the Kaye Validator AVS before starting this OQ test.

The Kaye Validator AVS instrument is designed to write data to the USB flash drive interface. To provide data security, data is stored in secure data files that can only be read by the Kaye Validator AVS software. Files that have been tampered with are no longer readable by the software.

Provide a freshly formatted USB flash drive for this test. The USB flash drive needs to be FAT32 formatted.

Note: The Kaye Validator AVS instrument reacts as no USB flash drive is connected when a bad or NTFS formatted flash drive is used!

Procedure:

1. Make sure that the USB flash drive is freshly FAT32 formatted.
2. Insert the USB flash drive in the Kaye Validator AVS instrument's USB Side port.
   Note: Wait at least 10 seconds after having connected the USB flash drive to the Kaye Validator AVS instrument's USB port before pressing the button to give the Kaye Validator AVS instrument enough time to recognize the flash drive.
3. Press the button for Data dump next to the USB Side port.
4. Wait until an audible Beep can be heard before removing USB flash drive. Record the results in Table 6.8.1.

Table 6.8.1. Data Recovery Verification Process with USB flash drive connected

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Sound	One audible Beep can be heard		
File(s) on USB flash drive	At least one **StudyFile_(***encrypted file name***).rep** can be found on the USB flash drive		

Comments:

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Emergency Data Recovery via USB Side port Functionality Verification (Section 6.8.)

Acceptance Criteria:

The Kaye Validator AVS instrument provides data recovery of the last studies by using the USB Side port and a common USB flash drive. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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6.9. File Management

Objective:
This test will verify that the Kaye Validator AVS system provides the possibility of Data Synchronization, Archiving, the data recovery of the last 10 studies using the Access Study File option and the ability to Convert the study files.

Note: OQ Section 6.5 should be performed with the Kaye Validator AVS system before starting this OQ test.

The Kaye Validator AVS instrument is designed to write data to the USB flash drive interface. To provide data security, data is stored in secure data files instrument that can only be read by the Kaye Validator AVS software. Files that have been tampered with, are no longer readable by the software.

Provide a freshly formatted USB flash drive for this test. The USB flash drive needs to be FAT32 formatted.

Note: The Kaye Validator AVS instrument reacts like no USB flash drive is connected when a bad or NTFS formatted flash drive is used!

6.9.1. Sync Out Functionality Verification

Procedure:

1. Make sure that the USB flash drive is freshly FAT32 formatted.
2. Connect a USB flash drive to the Kaye Validation Console. On the USB flash drive, create a new folder named "**Sync Out**." Record the results in Table 6.9.1.1.
3. Start the Kaye Validator AVS Software and create a new Asset with the name "**OQ Test Sync Out**". Enter all necessary information (Type: HeatBath) and Save the Asset.
4. Return to the **Activities** screen and select **File Management**.
5. Select **Sync Out** and login with a valid System Administrator.
6. The **Sync Out Location** window will appear.
7. Select **Browse**, navigate to the USB flash drive. (normally Drive D).
8. Select the **Sync Out** folder previously created and then **Select Folder.**
9. Select **OK** on the **Sync Out Location** window.
10. The **Sync Out Selections** window will appear.
11. Select **Users**, **Audit**, **Equipment** and **Templates** on the **Sync Out Selections** window. Also select the Asset **OQ Test Sync Out**. (A check will appear on top right corner) and select **OK**.
12. "Confirm Sync Out?" message appears – select **Yes**.
13. "Sync Out Successful" message will appear. Record the results in Table 6.9.1.1.
14. Use windows explorer to verify that the files and folders appear in the "Sync Out" folder on USB flash drive. Record the results in Table 6.9.1.1.

Table 6.9.1.1. Data Synchronization (Sync Out) Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Create Folder "Sync Out"	Folder created on FAT32 formatted USB flash drive		
Kaye Validation Console "Sync Out" confirmation	"Sync Out Successful." message appears after Sync Out process is complete.		
Asset **OQ Test Sync Out** information copied to the USB flash drive	Folders **AppData**, **Assets**, **Cal**, **Qual** and **Setups** are copied to the defined destination Folder. **Note:** An **EquipmentDetails** file will be also be copied if there are Equipment defined.		

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6.9.2. Archive Functionality Verification

Procedure:

1. Make sure that the USB flash drive is freshly FAT32 formatted.
2. Insert the USB flash drive in the Kaye Validation Console USB port. On the USB flash drive, create a new folder named "**Archive**." Record the results in Table 6.9.2.1.
3. Start the Kaye Validator AVS Software.
4. On the **Activities** screen and select **File Management**.
5. Select **Archive** and login with a valid System Administrator.
6. The **Archive Location and Date** window will appear.
7. Select **Browse**, navigate to the USB flash drive. (normally Drive D).
8. Select the **Archive** folder previously created then **Select Folder**.
9. Select an **Archive Date**(select current date) on the **Archive Location and Date** window. Select **OK**.
10. The **Archive Selections** window will appear.
11. Select **Setups**, **Study Files** and **Reports** on the **Archive Selections** window. Also select the Asset **OQ Tests**. (A check will appear on top right corner) and select **OK**.
12. "Confirm Archive?" message appears – select **Yes**.
13. "Archive Done Successfully. Please "Logout" and "Login" Application to get the refreshed data" message will appear. Record the results in Table 6.9.2.1.
14. Use windows explorer to verify that the files and folders appear in the "Archive" folder on USB flash drive. Record the results in Table 6.9.2.1.

Table 6.9.2.1. Archive Functionality Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Create Folder "Archive"	Folder created on FAT32 formatted USB flash drive		
Kaye Validation Console "Archive" confirmation	"Archive Done Successfully. Please "Logout" and "Login" Application to get the refreshed data" message appears after Archive process is complete.		
Asset **OQ Tests** information moved to the USB flash drive	Folders **Assets**, **Cal**, **Qual** and **Setups** are moved to the defined destination Folder		

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6.9.3. Sync In Functionality Verification (OQ Test Sync Out)

Procedure:

1. Start the Kaye Validator AVS Software and login with a valid System Administrator.
2. Delete the Asset **OQ Test Sync Out** located under the HeatBath column. Record the results in Table 6.9.3.1.
3. On the **Activities** screen select **File Management**.
4. Select **Sync In** and login with a valid System Administrator.
5. The **Sync In Location and Dates** window will appear.
6. Select **Browse**, navigate to the USB flash drive. (normally Drive D)
7. Select the **Sync Out** folder previously created in OQ Section 6.9.1 then **Select Folder**.
8. Uncheck **Filter By Date** on the **Sync In Location and Dates** window. Select **OK**.
   Note: If **Filter by date** is selected you'll have to choose a valid date.
9. The **Sync In Selections** window will appear.
10. Select **Users**, **Audit**, **Equipment** and **Templates** on the **Sync In Selections** window. Also select the Asset **OQ Test Sync Out** previously deleted. (A check will appear on top right corner) and select **OK**.
11. "Confirm Sync In?" window appears – select **Yes**.
12. "Sync In Successful. Please "Logout" and "Login" Application to get the refreshed data" message appears. Record the results in Table 6.9.3.1.
13. "Sign out" of the Application and Login to get the refreshed data. Record the Sync-In results in Table 6.9.3.1.

Table 6.9.3.1. Sync In Functionality Verification (OQ Test Sync Out)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Delete **OQ Test Sync Out** from the Software	**OQ Test Sync Out** no longer listed as an Asset		
Kaye Validation Console "Sync In" confirmation	"Sync In Successful. Please "Logout" and "Login" Application to get the refreshed data" message appears after Sync In process is complete.		
"Sync-In" of Asset **OQ Test Sync Out**	**OQ Test Sync Out** is again listed as an Asset		

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6.9.4. Sync In Functionality Verification (Archive)

Procedure:

1. Start the Kaye Validator AVS Software and login with a valid System Administrator.
2. Confirm **OQ Tests** has been moved and is no longer listed as an Asset. Record the results in Table 6.9.4.1.
3. On the **Activities** screen select **File Management**.
4. Select **Sync In** and login with a valid System Administrator.
5. The **Sync In Location and Dates** window will appear.
6. Select **Browse**, navigate to the USB flash drive. (normally Drive D)
7. Select the **Archive** folder previously created in OQ Section 6.9.2. then **Select Folder**.
8. Uncheck **Filter By Date** on the **Sync In Location and Dates** window. Select **OK**.
   Note: If **Filter by date** is selected you'll have to choose a valid date.
9. The **Sync In Selections** window will appear.
10. Select **Users**, **Audit, Equipment** and **Templates** on the **Sync In Selections** window. Also select the Asset **OQ Tests** previously moved. (A check will appear on top right corner) and select **OK**.
11. "Confirm Sync In?" window appears – select **Yes**.
12. "Sync In Successful. Please "Logout" and "Login" Application to get the refreshed data" message appears. Record the results in Table 6.9.4.1.
13. "Sign out" of the Application and Login to get the refreshed data. Record the Sync-In results in Table 6.9.4.1.

Table 6.9.4.1. Sync In Functionality Verification (Archive)

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Asset **OQ Tests** was moved	**OQ Tests** no longer listed as an Asset		
Kaye Validation Console "Sync In" confirmation	"Sync In Successful. Please "Logout" and "Login" Application to get the refreshed data" message appears after Sync In process is complete.		
"Sync-In" of Asset **OQ Tests**	**OQ Tests** is again listed as an Asset with all information for the Asset		

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6.9.5. Access Study Files Functionality Verification

Objective:
This test will verify that the Kaye Validator AVS system provides the possibility of uploading the last 10 Files from the Kaye Validator AVS instrument directly to Kaye Validation Console.

Note: Make sure that at least one or more Calibration, Qualification or Verification study have been performed with the Kaye Validator AVS system before starting this OQ test.

The Kaye Validator AVS instrument is designed to write data to the USB flash drive interface. To provide data security, data is stored in secure data files that can only be read by the Kaye Validator AVS software. Files that have been tampered with, are no longer readable by the software.

Procedure:

1. Start the Kaye Validator AVS Software and login with a valid System Administrator.
2. Select the Asset **OQ Tests** and delete the “OQ Functional Test Setup” Calibration, Qualification and Verification files. Record the results in Table 6.9.5.1.
3. Connect the Kaye Validation Console to the Kaye Validator AVS instrument.
4. Select **Hardware Connection**.
5. Select **Access Study Files**.
6. A list with the last 10 Study Files will appear.
7. Select the Calibration, Qualification and Verification associated with the “OQ Functional Test Setup” and select **Import Files**. Login with a valid User.
8. The selected files will be uploaded. Please “Logout” and “Login” Application to get the refreshed data” message appears. Record the results in Table 6.9.5.1.
9. Navigate to the appropriate **Asset** – **Calibrations**, **Qualifications** and **Verifications** and verify that the imported files appear. Record the results in Table 6.9.5.1

Table 6.9.5.1 Access Study Files Functionality Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Asset OQ Tests	The Calibration, Qualification and Verification files associated with “OQ Functional Test Setup” are deleted.		
Connect to Kaye Validator AVS	Live Data Screen is displayed		
Select **Hardware Connection** and **Access Study Files**	A list with the last 10 Study Files appear, Select Calibration, Qualification and Verification associated with the “OQ Functional Test Setup”		
Import Files and Login with a valid User	“Study File processed Successfully. Please “Logout” and “Login” Application to get the refreshed data” message appears after reading in the file		
Data Transfer verification	Asset OQ Test again contains the Calibration, Qualification and Verification Study files associated with “OQ Functional Test Setup”		

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6.9.6. AVS Convert Functionality Verification

Objective:

This test will verify the AVS Convert functionality of the Kaye Validator AVS system.

Procedure:

1. Connect the Kaye Validation Console to the Kaye Validator AVS instrument.
2. Select **Hardware Connection**, select **Access Study Files** and a list with the last 10 Study Files will appear.
3. Select all files other than the **OQ Test Setup 1** Qualification file.
4. Select **Delete Files** and login with a valid administrator account and confirm the deletion with **Yes** to remove these study files from the Kaye Validator AVS instrument's internal data storage. "Study Files Deleted Successfully" message will appear.
   Note: The "OQ Test Setup 1" Qualification file should be the only remaining study file in the internal data storage.
5. Insert a blank USB flash drive in the Kaye Validator AVS instrument's USB Side port.
6. Press the button for Data dump next to the USB Side port. Wait until an audible Beep can be heard before removing USB flash drive.
7. Insert the USB flash drive in the Kaye Validation Console. Or, if the Kaye Validation Console is docked insert in the USB port in the rear of the Kaye Validator AVS instrument with the picture of a Kaye Validation Console and verify that the .rep file has been written to the flash drive.
8. Start the Kaye Validator AVS Software and login with a valid System Administrator. Delete the **OQ Test Setup 1** Qualification study file from the Asset **OQ Tests**. Record the results in Table 6.9.6.1.
9. Select **File Management**. Select **AVS Convert** and enter a valid user.
10. Select **Browse** on the **Convert Raw File** window**.** Navigate to the USB flash drive. (normally Drive D)
11. File is shown with a page symbol, record the results in Table 6.9.6.1.
12. Select the file shown and select **Open.** The file name will populate in the **Convert Raw File** window.
13. Add a Comment: "Data Recovery Test" and select **OK**.
14. "Study File processed Successfully. Please "Logout" and "Login" Application to get the refreshed data" message appears. Record the results in Table 6.9.6.1.
15. "Sign out" of the Application and Login to get the refreshed data.
16. Navigate to the Asset **OQ Tests** and under **Qualifications** verify that the imported file with the added Comment appears. Record the results in Table 6.9.6.1.
17. When completed, return to the Activities screen.

Table 6.9.6.1. AVS Convert Functionality Verification

Verification Parameter	Specified Results	Results are as Specified Y/N	Initials/Date
Asset OQ Tests	The OQ Test Setup 1 Qualification study file is deleted.		
Data Transfer Procedure on Kaye Validation Console	File is shown with page symbol		
Kaye Validation Console "AVS Convert" confirmation	"Study File processed Successfully. Please "Logout" and "Login" Application to get the refreshed data" message appears after reading in the file		
Data Transfer verification	**OQ Test Setup 1** Qualification study file with added Comment "Data Recovery Test" is present		

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Comments:

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File Management (Sync Out / Archive / Sync In / Access Study Files / AVS Convert) Functionality Verification (Section 6.9.)

Acceptance Criteria:

The Kaye Validator AVS system provides Data Synchronization and Archiving (Sync Out / Archive / Sync In) functionality. The Kaye Validator AVS instrument provides data recovery of the last 10 studies using the Access Study File option and the ability to Convert a study file. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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7. Qualification Recovery

Objective:

The purpose of this procedure is to remove the Test Accounts from the Kaye Validator AVS System that were created for the purpose of executing this Protocol and to record the location of the Supporting Documentation.

Note: **Before deleting the Test Accounts, consider executing the Appendix A of this Protocol and the Kaye AVS Reporting Tool Operational Qualification Protocol!**

7.1. Deleting Test Accounts

Procedure:

1. Login to the Kaye Validator AVS software with your own Admin account. Select **Admin** and then **User Management**.
2. On the User Management screen select a Test Account from Table 7.1.1. and select Delete. Enter your own Admin account to confirm the deletion. Verify and document in Table 7.1.1. that the User Account has been deleted.
3. Repeat step 2 for all users listed in Table 7.1.1. Document your results in Table 7.1.1.
4. When all test accounts have been deleted, close the Software.

Table 7.1.1. Delete Test Accounts

User Name	User ID	User Type	Account Deleted Y/N	Initials/Date
Test System Administrator	2222	System Administrator		
Test Supervisor	3333	Supervisor		
Test Operator 1	4444	Operator		

7.2. Storage of Supporting Documentation

Procedure:

1. Record in Table 7.2.1. the location where all Supporting Documentation for the execution of this Protocol will be stored.

Table 7.2.1. Storage of Supporting Documentation

Location of Saved Supporting Documentation	Initials/Date

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Comments:

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Qualification Recovery (Section 7.)

Acceptance Criteria:

All Test Accounts have been deleted and verified. Storage location of the Supporting Documentation has been recorded and verified. The Acceptance Criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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Appendix A: SIM Calibration Information and Offsets Verification

Objective:

When a sensor calibration is completed, the Kaye Validator AVS system writes to the SIM the calibration setpoints, calibration date, offsets, the serial number of the Kaye Validator AVS instrument that was used to perform the calibration, and the SIM slot number to the individual SIM memory. This test will verify that each SIM properly stores the calibration information. This will be accomplished by performing a sensor calibration with each SIM and verifying that the calibration information is properly written to and stored in the SIM's memory.

Note: Performing a calibration with 1 Thermocouple connected is sufficient in verifying the SIM's ability to communicate and retain the calibration information in the SIM's memory. This procedure may be repeated for additional SIMs (e.g. newly purchased) by producing copies of the following tables and attaching them to this Protocol.

The calibration offsets are only valid when the SIM is in the same Kaye Validator AVS instrument, the same SIM slot number and when the same Setup, or a valid alteration of the original Setup is loaded in the Kaye Validator AVS instrument.

Procedure:

1. Using the "OQ Tests" Asset define a Setup "**SIM Calibration Setup**" with 4 sensors. **Under Sensors Configuration** configure channel #1 of all 4 SIM slots with the type of thermocouples available. (e.g. Type T TC (0.01)
2. Assign each of the Sensors a Label of "**Temp**" and select **Yes** when software asks, "Is it intentional?".
3. Under **Group Sensors** use Default Group to assign all sensors in one Group.
4. Select **Calculations** and keep the default calculations.
5. Select **Cal/Ver Parameters**. Keep the default Setpoints and enter the Stability criteria of Sensors for 3 Minutes, IRTD for 1 Minute. Under the **"Calibration Temperature Setpoints",** uncheck the **Check** setpoint.
6. Select **Qualification Parameters** and keep the default parameters.
7. Select **Review** and save the Setup.
8. Wire the SIM(s) that will be used with the Kaye Validator AVS system with at least 1 thermocouple connected to channel 1 of the SIM(s).
9. Connect the SIM(s) to the Kaye Validator AVS instrument. Record the SIM(s) serial number(s) in the corresponding Tables on the following pages.
10. Perform a calibration using the "**SIM Calibration Setup**".

    Note: Under **Select Sensors for Calibration**, you may select to calibrate only the location(s) that have sensor(s) connected.
11. Once the calibration is complete, navigate to the Hardware Connection screen (refer to section 6.3.), select Sensor Offsets and record the calibration information displayed for each SIM in the corresponding Tables.
12. Close Sensor Offset screen and return to the Activities screen.
13. Select "**Connected**" on Activities screen to disconnect the Kaye Validation Console from the Kaye Validator AVS instrument.
14. Power down the Kaye Validator AVS instrument and wait until Power LED is off.
15. Move the SIM(s) from the slot(s) in which it was calibrated to a different slot(s).
16. Power up the Kaye Validator AVS instrument and wait till booting has been finished (Power LED permanently green).

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17. Connect the Kaye Validation Console to the Kaye Validator AVS instrument and navigate to Hardware – Sensor Offset screen and verify that the sensor for the SIM(s) displays as “Uncalibrated.” Record the results in the corresponding Tables.
18. Close Sensor Offset screen and return to the Activities screen.
19. Select “**Connected**” on Activities screen to disconnect the Kaye Validation Console from the Kaye Validator AVS. Close the Kaye Validator AVS Software.
20. Power down the Kaye Validator AVS instrument and return the SIM(s) to the original slot in which the calibration was performed.
21. Power up the Kaye Validator AVS instrument and wait until booting has been finished (Power LED permanently green).
22. Connect the Kaye Validation Console to the Kaye Validator AVS instrument and navigate to Hardware – Sensor Offset screen and verify that the offsets are the same as those recorded in step 11. Record the results in the corresponding Tables.
23. Close Sensor Offset screen and return to the Activities screen.

SIM 1 Serial Number: _____

SIM Calibrated In Slot	SIM Calibrated In AVS S/N	Low Calibration Point	High Calibration Point	Check Point	Sensor Calibration Date	Information stored in SIM Y/N	Initials/Date

SIM Channel	Input Min	Output Min	Input Max	Output Max	"Uncalibrated" Displayed after SIMs Moved Y/N	Correct Offsets displayed after returning SIM to original Slot Y/N	Initials/Date
1							

SIM 2 Serial Number: _____

SIM Calibrated In Slot	SIM Calibrated In AVS S/N	Low Calibration Point	High Calibration Point	Check Point	Sensor Calibration Date	Information stored in SIM Y/N	Initials/Date

SIM Channel	Input Min	Output Min	Input Max	Output Max	"Uncalibrated" Displayed after SIMs Moved Y/N	Correct Offsets displayed after returning SIM to original Slot Y/N	Initials/Date
1							

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(Print additional copies of this page as needed) Page _____ of _____ .

SIM_____ Serial Number: _____

SIM Calibrated In Slot	SIM Calibrated In AVS S/N	Low Calibration Point	High Calibration Point	Check Point	Sensor Calibration Date	Information stored in SIM Y/N	Initials/Date

SIM Channel	Input Min	Output Min	Input Max	Output Max	"Uncalibrated" Displayed after SIMs Moved Y/N	Correct Offsets displayed after returning SIM to original Slot Y/N	Initials/Date
1							

SIM_____ Serial Number: _____

SIM Calibrated In Slot	SIM Calibrated In AVS S/N	Low Calibration Point	High Calibration Point	Check Point	Sensor Calibration Date	Information stored in SIM Y/N	Initials/Date

SIM Channel	Input Min	Output Min	Input Max	Output Max	"Uncalibrated" Displayed after SIMs Moved Y/N	Correct Offsets displayed after returning SIM to original Slot Y/N	Initials/Date
1							

SIM_____ Serial Number: _____

SIM Calibrated In Slot	SIM Calibrated In AVS S/N	Low Calibration Point	High Calibration Point	Check Point	Sensor Calibration Date	Information stored in SIM Y/N	Initials/Date

SIM Channel	Input Min	Output Min	Input Max	Output Max	"Uncalibrated" Displayed after SIMs Moved Y/N	Correct Offsets displayed after returning SIM to original Slot Y/N	Initials/Date
1							

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Comments:

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Appendix A: SIM Calibration Information and Offsets Verification

Acceptance Criteria:

The Kaye Validator AVS software recognizes the calibration information and offsets stored in each SIM that will be used with the Kaye Validator AVS system. The acceptance criteria for this section are derived from Amphenol Advanced Sensors recommendations.

All Acceptance Criteria have been met. Yes ____ No ____ (If No, explain in Comments)

Reviewed By: _____ Date: _____

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8. DISCREPANCY REPORT

Procedure:

1. Assign a sequential identification number to the Discrepancy and record the Protocol section or page number that the Discrepancy references.
2. Document a complete description of the Discrepancy, specifically addressing how it impacts the Acceptance Criteria of the Validation procedure. Sign and date the Discrepancy description and notify the designated reviewer.
3. The responsible personnel must determine a suitable Resolution to the Discrepancy and document the planned Resolution on the Discrepancy Report.
4. The designated reviewer must verify that the Resolution has been executed satisfactorily and sign and date the Discrepancy Report.
5. Submit the Discrepancy Report to the designated QA reviewer for approval. Include the approved Discrepancy Report as Supporting Documentation for the affected Validation procedure.

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8. DISCREPANCY REPORT (continued)

Discrepancy Report Form
(Copy this form as needed) Page _____ of _____ .

Discrepancy No.:		**Protocol Section:**		**Page No.:**	
Validation Discrepancy Description:					
Acceptance Criteria Impact:					
Submitted By:				**Date:**	
Validation Discrepancy Resolution:					
Recorded By:				**Date:**	
Reviewed By:				**Date:**	
QA Approval:				**Date:**	

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